

Exponential towers

(OEIS A096537 and A096542)

Airy limits and the corrected asymptotic of A096537;
fractional corrections and Gamma laws for the shifted family

Two research reports of one external research session
(bundle Reports 192 and 193), merged into one ProveIt report

Both Parts: 4 October 2026 merged: 7 October 2026

Abstract

Let $F_1(x) = \exp(x \exp(2x \exp(3x \cdots))) = \sum a_n x^n / n!$ (OEIS A096537), $F_y = \exp(xy F_{y+1})$ the shifted family whose coefficients $P_n(y)$ are the rows of A096542, and $\rho = 1 - e^{-1}$. This report prints two manuscripts, a theorem and its sequel. Part I proves $a_n / (n!)^2 \sim \sqrt{2/(\pi\rho)} \rho^{-n} n^{-1/2}$, that is $a_n \sim 2\sqrt{2\pi/\rho} n^{2n+1/2} e^{-n} (e-1)^{-n}$, through the critical finite-height towers $H_h(1/(eh)) \sim Kh^{1/6} e^{-(e-1)h}$, an Airy size law, positive-factorization minor arcs and a uniform lattice local limit; it also proves a Gaussian height law for depth-weighted labelled trees and a refined Lambert inverse. Part II proves the first correction $a_n \rho^n \sqrt{n} / (n!)^2 = C + c_1 n^{-1/3} + o(n^{-1/3})$ with $C = \sqrt{2/(\pi\rho)}$ and $-0.61 < c_1 < -0.25$, c_1 an absolutely convergent Airy integral whose sign is certified by a real reduction and directed interval quadrature, and the Gamma prefactor $P_n(y)/(n!)^2 \sim C \rho^{1-y} \rho^{-n} n^{y-3/2} / \Gamma(y)$ for every fixed $y > 0$, which answers two of Part I's open questions. The write records two refuted OEIS formulas, with proofs: the amplitude 2π conjectured by Václav Kotěšovec in A096537 is wrong (the correct amplitude is larger by the factor $\sqrt{2/(\pi\rho)} = 1.00355\dots$), and the A096542 formula $T(n, 1) = n a_n$ should read $T(n, 1) = n a_{n-1}$. Further coefficients, effective constants and onsets, extreme shifts and a complex Borel continuation remain open. The report is unrefereed, and nothing in it is formalized.

Contents

Guide to this report	4
What is proved, and where	5
The OEIS entries	5
The inverses and the transseries volume	6
Notation across the two Parts	6
Provenance and merge decisions	7
What was checked, and what was not	7
Relation to neighbouring reports and formal projects	8

I	Airy limits and the corrected asymptotic of A096537: critical exponential towers and depth weighted labeled trees	9
1	Results and proof structure	9
2	Exact coefficients and weighted trees	12
3	Critical entrance and sensitivity	13
4	Bulk matching and the endpoint product	15
5	Critical finite-height amplitude	17
6	A uniform real critical window	18
7	The Airy size law and its density	21
8	Fourier minor arcs from positive factorization	22
9	Uniform lattice transfer	25
10	All-height domination and the coefficient sum	25
11	Gaussian height fluctuations	27
12	Refined inversion and the integer threshold	27
13	Exact verification and reproducibility	28
14	Prior sources and remaining questions	30
II	Fractional corrections and Gamma laws for exponential towers: a sequel to the leading Airy asymptotic	31
15	Statements and normalizations	32
16	Exact heights and the leading input	34
17	The normalized parabolic entrance	35
18	Complex stability and quantitative inner propagation	37
19	Second outer correction and overlap cancellation	39
20	The Gamma factor on an offset grid	41
21	Fourier tails and the full first order remainder	43
22	Height averaging and the universal relative correction	45
23	A real Airy integral for the correction	46
24	An explicit tail bound and a certified sign	50
25	Ancestor choices and local Gamma laws	53
26	The positive real Borel boundary	55

27	Threshold inversion at the fractional scale	55
28	Reproducibility and source scope	56
29	Further questions and research	58

Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 8622ca7e5 (batch 110, cluster 110-TOWER).

Part	Bundle report	Delivered title	Labels
I	192 (base)	Airy Limits and the Corrected Asymptotic of A096537: Critical exponential towers and depth weighted labeled trees (4 October 2026)	xtw:airy:
II	193	Fractional Corrections and Gamma Laws for Exponential Towers: A sequel to the leading Airy asymptotic (4 October 2026)	xtw:frac:

One object, a theorem and its sequel. Both Parts study Hanna’s continued exponential $F_1(x) = \exp(x \exp(2x \exp(3x \cdots)))$, the coefficients $a_n = n![x^n]F_1$ of OEIS A096537, and the shifted family $F_y = \exp(xyF_{y+1})$ whose coefficients $P_n(y)$ are the rows of A096542, with the same constants $\rho = 1 - e^{-1}$, $\alpha = e - 1$ and the same weighted labelled trees. Part I proves the leading equivalent $a_n/(n!)^2 \sim \sqrt{2/(\pi\rho)} \rho^{-n} n^{-1/2}$ through critical finite-height towers, an Airy size law and a uniform lattice local limit; this **refutes the amplitude 2π conjectured in A096537** (Remark 1.4). Part II is an explicit sequel. It cites Part I for the leading theorem, restates the leading facts it uses (Section 16), and answers two items of Part I’s list of open questions: it proves the first correction, $b_n \rho^n \sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3})$ with $-0.61 < c_1 < -0.25$ (Theorem 15.1), and the Gamma prefactor for every fixed positive shift (Theorem 15.2). At $y = 1$ its Theorem 15.2 is Part I’s Theorem 1.1 with the correction added, and its inverse (222) is Part I’s $r(Y)$ of (14) with one further term.

Arrangement. The base of the merge is Report 192; its `Report192.tex` was staged as this `article.tex`, and it is printed first, as Part I, because Report 193 depends on it. Report 193 is printed second, as Part II, in full: its proofs of the leading facts are part of a quantitative argument that also gives the correction, so nothing is removed as a duplicate; where it reproves a statement of Part I, a dated note says so.

Numbering. Sections are numbered continuously; statements are numbered within sections and equations continuously through the report, as in both manuscripts. Part I keeps Report 192’s Sections 1–14, its equations (1)–(109) and all its other numbers; the write’s two remarks (Remarks 1.4 and 13.1) are placed after the last numbered statement of their sections, so that no delivered number moves. Report 193’s Section k is Section $k + 14$ here (Sections 15–28), its statements move with their sections (Report 193’s Theorem 1.1 is Theorem 15.1), and its equation (k) is equation $(k + 109)$. Section 29, which gathers the open questions of both Parts under Vladimir’s standing rule of 4 October 2026, was added in the write.

Notes added in the write are set in small type and tagged [**write**], with their date (7 October 2026). No delivered statement was changed and no delivered symbol was renamed.

What is proved, and where

Statement	Where
<i>The leading theory (Part I)</i>	
Exact-height coefficients and depth-weighted labelled trees;	Section 2, Lemma 2.1
$a_n \geq n^2 a_{n-1}$	
$b_n = a_n / (n!)^2 \sim \sqrt{2/(\pi\rho)} \rho^{-n} n^{-1/2}$, i.e.	Theorem 1.1, Remark 1.4
$a_n \sim 2\sqrt{2\pi/\rho} n^{2n+1/2} e^{-n} (e-1)^{-n}$; the OEIS amplitude 2π is refuted	
Critical tower $H_h(1/(eh)) \sim Kh^{1/6} e^{-\alpha h}$; uniform real Airy window	Theorems 5.1 and 6.1
Airy size law with $p(0) = \text{Ai}(0)^2/\beta$; uniform minor arcs; uniform lattice local limit	Theorems 1.2, 8.1, Section 9
Gaussian height limit of depth-weighted trees	Section 11
Refined Lambert inverse and shrinking integer sandwich	Theorem 1.3, Section 12
$T(n, 1) = n a_{n-1}$ in A096542; the entry's formula	Section 13, Remark 13.1
$T(n, 1) = n a_n$ is refuted	
<i>The sequel (Part II)</i>	
$b_n \rho^n \sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3})$, $-0.61 < c_1 < -0.25$, c_1 an absolutely convergent Airy integral	Theorems 15.1, 23.1, Section 24
$b_n(y) \sim C \rho^{1-y} \rho^{-n} n^{y-3/2} / \Gamma(y)$ with the same relative correction, uniformly on compact $Y \subset (0, \infty)$; normal limit of root-ancestor choices	Theorem 15.2, Section 25
Local Gamma correction; positive real Borel boundary; threshold inverses at the fractional scale	Theorem 25.1, Sections 26 and 27
<i>Open</i>	
The next coefficient scale; effective rates, onsets and certified inversion; sharper certified constants; shifts near 0 or growing; larger tilt domains and rare heights; a complex Borel continuation; the full comparison with Bender–Vinson and Poland. See Section 29.	

The finite computations (exact integers, rational checks, and Part II's directed interval quadrature for the sign of c_1) check identities, normalizations and one finite enclosure; no asymptotic statement rests on them, and the interval certificate rests on the analytic reduction and tail bound proved in Part II.

The OEIS entries

The write read both entries on 7 October 2026. **A096537** [1] (revision #8 of 17 December 2019), by Paul D. Hanna (24 June 2004), has the formula field “Conjecture: $a(n) \sim 2 * \text{Pi} * n^{(2*n + 1/2)} / (\exp(n) * (\exp(1) - 1)^n)$. - Vaclav Kotesovec, Dec 16 2019”. This conjecture is **false**: Part I's Theorem 1.1 gives the same expression with amplitude $2\sqrt{2\pi/\rho}$, and the two amplitudes differ because $\pi\rho < 2$ (Remark 1.4). The entry's other formula, “ $a(n) = \text{A096542}(n+1, 1)/(n+1)$ ”, is right. **A096542** [2] (revision #11 of 19 December 2017), by Hanna (25 June 2004), has the formula line “ $T(n, 1) = n * \text{A096537}(n)$ ”. This line is **false** at every $n \geq 2$; the correct statement is $T(n, 1) = n \text{A096537}(n-1)$, and $n=2$ is a counterexample: the entry's own row 2 is 0, 2, 3, so $T(2, 1) = 2 = 2 \cdot \text{A096537}(1)$, while $2 \cdot \text{A096537}(2) = 10$ (Remark 13.1). Both manuscripts record this index mismatch (Sections 13 and 28). Nothing was submitted to the OEIS; whether to propose the corrections is a decision for a human editor. The OEIS terms in `data/192-airy-oeis_fixtures.json` are OEIS content under CC BY-SA 4.0; `data/192-airy-generated_coefficients_48.json`

is generated by the package, not taken from the OEIS.

The inverses and the transseries volume

The repository’s transseries volume [12] has a Part 0 apparatus for inverting rapidly growing functions; its symbols are subscripted “vol” here. Statement by statement:

- Part I’s (108), $N(Y) = \lceil z(Y) \rceil$ with z the inverse of the piecewise-linear interpolation g of $\log a_n$, and Part II’s (218), the same for $\log P_n(y)$: instances of item (1) of `p0:thm:staircase`, with $A_n = \log a_n$ (resp. $\log P_n(y)$) on its eventually increasing part and the admissible interpolation g (resp. g_y). The sandwiches (16) and (223) follow from item (1) and the monotonicity of the ceiling, as item (2) does; the real-inverse estimates that feed them are the Parts’ own.
- The Lambert base $x = L/(2W_0(L/(2e\sqrt{\rho})))$ of Part I (14) and Part II (220) (where it is X), the root of $f(t) = 2t \log(t/(e\sqrt{\rho})) = L$: an *instance* of the factorial core `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 2$, $d_{\text{vol}} = -(2 + \log \rho)$ and target L , which gives $v_{\text{vol}} = W_0(L/(2e\sqrt{\rho}))$ and $X_{\text{vol}} = e\sqrt{\rho} e^{v_{\text{vol}}} = L/(2v_{\text{vol}})$.
- The refined inverses (107), (219), (221) and (222): first-order (Newton) terms of the volume’s perturbation-operator series `p0:eq:operator-series` about the core f , with error bounds proved by the Parts; *analogues*, not instances of a theorem of the volume.

Neither Part claims novelty for the inversion mechanics.

Notation across the two Parts

Common to both Parts with the same meaning: F_y , $P_n(y)$, a_n , b_n ($= b_n(1)$ in Part II), ρ , α , $c_{n,h}$, H_h , Ai, Bi, and the Riccati objects w , U (Part II’s w_λ , U_λ reduce to Part I’s at $\lambda = 0$). No symbol was renamed. The letters below change meaning:

Symbol	Meanings
β , a	Part I: $\beta = 2^{-1/3}$. Part II: $a = 2^{-1/3}$ is that constant, and $\beta = y - 1$ is the shift offset (Section 16). <i>False reading</i> : Part II’s β is not $2^{-1/3}$.
K , κ	Part I: $K = 2^{1/6} \sqrt{e}/(\sqrt{\pi} \text{Ai}(0)^2)$ the critical constant. Part II: $\kappa = 2^{1/6} \sqrt{e/\pi}$, so $K = \kappa/\text{Ai}(0)^2 = \mathcal{F}(0)$; K an integer cutoff (131); K_n the root-ancestor statistic.
C , c_1 , d	Part I writes $\sqrt{2/(\pi\rho)}$ without a name. Part II: $C = \sqrt{2/(\pi\rho)}$, c_1 the first correction, $d = c_1/C$; d_y a Newton displacement. Part I: no C .
ε , m	Part I: $\varepsilon = 1/(eh)$ the mesh, $m = h^{1/3}$. Part II: $\epsilon = h^{-1/3} = 1/m$. <i>False reading</i> : the two epsilons are different scales.
p , f	Part I: p the Airy size density (74); $f(t) = 2t \log(t/(e\sqrt{\rho}))$ in Section 12. Part II: $f(z)$ in Section 16 is $Kp(z)$, Part I’s density times the critical constant, since $\mathcal{F}(it) = K\psi(t)$; so $f(0) = Kp(0) = \kappa/a$; f is the same core as Part I’s in Section 27. <i>False reading</i> : Part II’s f is not a probability density.
x , X , r , z	Part I: x the Lambert base, $r(Y)$ the refined inverse, $z(Y)$ the interpolated inverse. Part II: X is that base, r_y a Newton point, $r(\lambda) = \text{Bi}/\text{Ai}$, $z_y(L)$ the interpolated inverse.
L_n , N	Part I: L_n the height of the weighted tree; N_h the size variable; $N(Y)$ the threshold. Part II: $L_n = \log(n/\rho)$; $N_y(Y_0)$ the threshold.
T , D , M , J , Q	Part I: $T(z)$ the tree function, D_i differences or factors, J a divided difference or a compact set, Q_j approximants. Part II: $T = \epsilon M$ and M cutoffs, $D_c(t)$, $D = 2 \log(X/\sqrt{\rho})$, $J_c(\lambda)$, $Q(\lambda)$ the correction integrand, $M_y(t)$ the smooth log model.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 4 October 2026. Their title blocks read “Report192” and “Report193”, with the subtitles above in the author slot; both PDF author fields read “Research report”; neither names an author, a tool or an addressee, neither pins a repository commit, and neither carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by 8622ca7e5 (batch 110, cluster 110-TOWER).

- **Part I**, bundle Report 192, archive A096537_Corrected_Asymptotic_and_Airy_Limits_Source.zip (662,864 bytes; 23 files; Report192.tex, 1,502 lines, 24 pages); the base.
- **Part II**, bundle Report 193, archive A096537_Fractional_Corrections_and_Gamma_Laws_Source.zip (1,008,394 bytes; 37 files; Report193.tex, 1,965 lines, 29 pages). It cites Report 192 as report192; that bibliography entry now points to Part I. Three of its generated files, generated/certificate/coefficient.json, core.json and panels.json, are byte copies of its fixtures code/fixtures/*_reference.json and are not shipped.

Where the merge had to choose:

1. *Base and order*. Report 192 first (see the Guide).
2. *Repeated arguments*. Part II re-derives the leading facts it uses inside a quantitative argument (the parabolic entrance with the forced logarithm, the positive factorization on minor arcs, the height averaging); each proof is needed for its own theorem, so both are printed in full, with dated notes in Part II naming the Part I counterparts.
3. *Front matter, numbering and bibliography*. The two title blocks and abstracts are printed at the heads of their Parts, with each manuscript’s opening paragraph; one table of contents replaces two. The two bibliographies are merged: the seven shared keys (oeis537, oeis542, bv, poland, dlmf2, dlmf7, dlmf9) are single entries keeping both Parts’ details, report192 points to Part I, and the write added the entry for the transseries volume.
4. *Write additions*. This front matter, Remarks 1.4 and 13.1, Section 29, the dated notes and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 110, 6 October 2026) both manuscripts were read and no mathematical error was found. An independent computation of the full-height tower coefficients reproduced 1, 1, 5, 61, 1377, 49721, 2625313, 190735749, 18246616321 and gave $b_n \rho^n \sqrt{n} = 0.9144, 0.9462, 0.9609, 0.9695$ at $n = 100, 400, 1000, 2000$; least-squares fits $c_0 + \sum_j c_j n^{-j/3}$ on the ranges $n \geq 200, 400, 800$ gave c_0 between 1.00356 and 1.00363, against $\sqrt{2}/(\pi\rho) = 1.0035525\dots$ (the conjectured amplitude corresponds to $c_0 = 1$), and $c_1 \approx -0.437$, inside Part II’s enclosure; exact rows of A096542 through $n = 8$ confirmed $T(n, 1) = n a_{n-1}$. These are numerical observations, not proofs. The delivered suites were rerun on copies (results in the README). The write (7 October 2026) read both OEIS entries, checked the inequality $\pi\rho < 2$ and the rows of A096542 quoted in Remark 13.1, checked the identifications with the transseries volume above, and reran Part I’s exact checker and Part II’s exact checker and interval tests on copies. What was *not* done: the analytic proofs were read and their displayed algebra spot-checked, but they are the manuscripts’ proofs, not an independent verification; Bender–Vinson, Poland and

Chapoton were not read by the write; Part II's 562-panel interval certificate was rerun by the intake but not by the write.

Relation to neighbouring reports and formal projects

No other placed report treats A096537, A096542, continued exponentials or depth-weighted labelled trees (repository search, 6–7 October 2026). The transseries volume [12] is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

Airy limits and the corrected asymptotic of A096537: critical exponential towers and depth weighted labeled trees

Bundle Report 192, archive A096537_Corrected_Asymptotic_and_Airy_Limits_Source.zip, delivered as Report192.tex (24 pages), the base of the merge. Delivered title block “Report192 / Airy Limits and the Corrected Asymptotic of A096537 / Critical exponential towers and depth weighted labeled trees / 4 October 2026”. Printed as delivered, with [write] notes and two [write] remarks; section, statement and equation numbers are those of the delivery. The opening paragraph “Attribution and scope” is the manuscript’s own.

Abstract of this Part (as delivered). For the formal exponential generating function $F_1(x) = \exp(x \exp(2x \exp(3x \cdots))) = \sum a_n x^n / n!$, we prove

$$\frac{a_n}{(n!)^2} \sim \sqrt{\frac{2}{\pi\rho}} \rho^{-n} n^{-1/2}, \quad \rho = 1 - e^{-1}.$$

Thus the amplitude 2π in the current OEIS conjecture must be replaced by $2\sqrt{2\pi/\rho}$. The ratio is 1.00355251532641107955... The proof derives the critical finite-height tower equivalent, including its $h^{1/6}$ amplitude, from an entrance Riccati equation and a normalized Airy solution. A uniform real tilt window yields an Airy size law; a separate positive factorization controls every Fourier minor arc and gives a uniform lattice local limit. An all-height bound then justifies the final Laplace sum. We also prove a Gaussian height limit for the associated depth weighted labeled trees and a refined Lambert-function inverse with a shrinking integer sandwich. The report supplies self-contained proofs of these steps and exact reproducible finite checks. No effective asymptotic error, all-orders expansion, or claim of worldwide priority is made.

Attribution and scope. The sequence and the entire shifted functional equation are prior constructions of Paul D. Hanna recorded in OEIS A096537 and A096542 [1, 2]. General continued exponentials and their weighted tree interpretation are part of the earlier framework of Bender and Vinson [3]. The exact theorem scope of that paper and of Poland’s continuation [4] remains unresolved in the available full-text comparison. We prove the formulas stated here; we do not infer their novelty from unsuccessful searches. Section 14 records the precise limits of the comparison.

1 Results and proof structure

All logarithms are natural. Set

$$\alpha = e - 1, \quad \rho = \alpha/e = 1 - e^{-1}, \quad \beta = 2^{-1/3}, \quad K = \frac{2^{1/6}\sqrt{e}}{\sqrt{\pi} \operatorname{Ai}(0)^2}. \quad (1)$$

The Airy functions have their usual normalization as solutions of $y'' = zy$. In particular $\operatorname{Ai}(0) = 1/(3^{2/3}\Gamma(2/3))$. The functional equations

$$F_y(x) = \exp(xyF_{y+1}(x)), \quad F_y(0) = 1, \quad (2)$$

define a unique family of formal series: the coefficient of degree n on the right uses coefficients only through degree $n - 1$ of the next series. We put

$$a_n = n![x^n]F_1(x), \quad b_n = \frac{a_n}{(n!)^2}. \quad (3)$$

No analytic convergence of the infinite tower is assumed or used. Every analytic evaluation in the proof is of a finite entire tower.

Theorem 1.1 (Corrected coefficient equivalent). *As $n \rightarrow \infty$,*

$$b_n \sim \sqrt{\frac{2}{\pi\rho}} \rho^{-n} n^{-1/2}, \quad (4)$$

$$a_n \sim 2\sqrt{\frac{2\pi}{\rho}} \frac{n^{2n+1/2}}{e^n(e-1)^n}. \quad (5)$$

Equivalently,

$$\log a_n = 2n \log n - (1 + \log(e-1))n + \frac{1}{2} \log n + \log\left(2\sqrt{\frac{2\pi}{\rho}}\right) + o(1). \quad (6)$$

The retrieved A096537 entry attributes to Václav Kotešovec a conjecture, dated 16 December 2019, with the same expression as (5) but amplitude 2π [1]. The corrected-to-conjectured ratio is

$$\sqrt{\frac{2}{\pi\rho}} = 1.00355251532641107955\dots > 1. \quad (7)$$

The strict inequality does not depend on decimal computation: $e < 11/4$ gives $\rho < 7/11$, whereas $\pi < 22/7$, so $\pi\rho < 2$. Thus Theorem 1.1 disproves that particular amplitude conjecture.

For $h \geq 1$, define

$$A_{h+1,h}(x) = 1, \quad A_{i,h}(x) = \exp(ixA_{i+1,h}(x)) \quad (1 \leq i \leq h), \quad (8)$$

and write $A_{1,0} = 1$ and $H_h = A_{1,h} - A_{1,h-1}$. These entire functions have nonnegative Taylor coefficients, with H_h starting in degree h . Theorem 6.1 below gives, uniformly for λ in each compact subset of $(-\infty, 1]$,

$$H_h\left(\frac{1 + \lambda h^{-2/3}}{eh}\right) \sim \frac{2^{1/6}\sqrt{e}}{\sqrt{\pi} \operatorname{Ai}(-\beta\lambda)^2} h^{1/6} e^{-\alpha h + \alpha\lambda h^{1/3}}. \quad (9)$$

The upper endpoint 1 is a convenient sufficient restriction, not an assertion about the largest possible parameter domain.

Theorem 1.2 (Two different fluctuation laws). *Define $c_{n,h} = [x^n]H_h(x)$ and an integer variable N_h by*

$$\mathbb{P}(N_h = n) = \frac{c_{n,h}(eh)^{-n}}{H_h(1/(eh))}. \quad (10)$$

Then $(N_h - \alpha h)/h^{2/3}$ converges to a law with bounded continuous density p and characteristic function

$$\psi(t) = \frac{\operatorname{Ai}(0)^2}{\operatorname{Ai}(-\beta it)^2}, \quad p(0) = \frac{\operatorname{Ai}(0)^2}{\beta}. \quad (11)$$

Moreover,

$$\sup_{n \in \mathbb{Z}} \left| h^{2/3} \mathbb{P}(N_h = n) - p\left(\frac{n - \alpha h}{h^{2/3}}\right) \right| \longrightarrow 0. \quad (12)$$

Separately, let L_n be the height of a labeled tree on $\{0, \dots, n\}$, rooted at 0, chosen with probability proportional to the product of all nonroot depths. Then

$$\frac{L_n - n/\alpha}{\sqrt{n}/\alpha} \xrightarrow{d} \mathcal{N}(0, 1). \quad (13)$$

The finite-height amplitude $h^{1/6}$, the size fluctuation scale $h^{2/3}$, and the fixed-size height fluctuation scale $\sqrt{n}$ describe different objects. They must not be identified with one another.

Theorem 1.3 (Refined threshold inverse). *Let $Y \rightarrow \infty$, $L = \log Y$, and*

$$x = \frac{L}{2W_0(L/(2e\sqrt{\rho}))}, \quad r(Y) = x - \frac{1}{4} - \frac{\log(8\pi/\sqrt{\rho})}{4\log(x/\sqrt{\rho})}. \quad (14)$$

Here W_0 is the positive real Lambert function. The inverse $z(Y)$ of the continuous piecewise-linear interpolation of $\log a_n$, on its eventually strictly increasing portion, satisfies

$$z(Y) = r(Y) + o(1/\log x). \quad (15)$$

For $N(Y) = \min\{n : a_n \geq Y\}$ there exists $\epsilon(Y) > 0$ tending to zero such that, for all sufficiently large Y ,

$$\left\lceil r(Y) - \frac{\epsilon(Y)}{\log x} \right\rceil \leq N(Y) \leq \left\lceil r(Y) + \frac{\epsilon(Y)}{\log x} \right\rceil. \quad (16)$$

This is an existence statement for a vanishing error envelope, not an effective finite-target bound or a universally valid formula $N(Y) = \lceil r(Y) \rceil$.

1.1 Dependencies and the role of each argument

The proof has the following order.

1. Section 2 identifies the exact-height coefficients and the weighted-tree normalization, including the two factorials in b_n .
2. Sections 3–5 derive the critical entrance and its sensitivity, prove bulk product matching, and obtain K .
3. Sections 6–7 establish the uniform real window, complex characteristic-function limit by analytic continuation, and $p(0)$ by a Wronskian integral.
4. Section 8 proves Fourier-tail bounds independently of the Airy approximation. Section 9 then proves the uniform lattice local limit. Weak convergence alone would not suffice here.
5. Section 10 obtains an all-height coefficient majorant and performs the height sum. This is where the Airy constants cancel.
6. Sections 11 and 12 derive the height central limit theorem and the refined inverse.

Every limiting matching argument first takes $h \rightarrow \infty$ at a fixed positive matching time, then sends that time to infinity. No unstated uniformity along a growing matching time is used. Exact computations in Section 13 check identities and normalizations; they do not replace any asymptotic proof.

Remark 1.4 ([write] The amplitude conjectured in A096537 is false). [Added 7 October 2026, batch 110.] The formula field of OEIS A096537 (revision #8, 17 December 2019, live on 7 October 2026) reads “Conjecture: $a(n) \sim 2 * \text{Pi} * n^{(2*n + 1/2)} / (\exp(n) * (\exp(1) - 1)^n)$. - Vaclav Kotesovec, Dec 16 2019”. This conjecture is false. By Theorem 1.1,

$$\frac{a_n e^n (e-1)^n}{n^{2n+1/2}} \longrightarrow 2\sqrt{\frac{2\pi}{\rho}} = 2\pi\sqrt{\frac{2}{\pi\rho}},$$

whereas the conjecture asserts the limit 2π . The two limits differ because $\pi\rho < 2$: from $e < 11/4$, $\rho = 1 - 1/e < 1 - 4/11 = 7/11$, and $\pi < 22/7$, so $\pi\rho < 2$, as shown after (7). So the conjectured form $n^{2n+1/2}e^{-n}(e-1)^{-n}$ is right and only its amplitude is wrong, by the factor $\sqrt{2/(\pi\rho)} = 1.00355251532641107954\dots$ (the abstract and (7) print $\dots 07955\dots$, a rounding; the truncation is $\dots 07954$). The intake's independent computation of a_n to $n = 2000$, with fits in powers of $n^{-1/3}$, extrapolated $a_n\rho^n\sqrt{n}/(n!)^2$ to 1.0036, not to 1. The approach is slow because of the first correction $c_1n^{-1/3}$ proved in Part II (Theorem 15.1): with Part II's diagnostic value $c_1 \approx -0.4365$, the normalized value at $n = 2000$ is about 0.969, about 3.5 % below its limit, which is why a short-range fit could not have distinguished the two amplitudes. No correction has been submitted to the OEIS.

2 Exact coefficients and weighted trees

Let $\mathcal{T}_n$ be the trees with vertex set $\{0, 1, \dots, n\}$ and root 0. For $T \in \mathcal{T}_n$, let $d_T(v)$ be the distance from 0 to v and set

$$w_y(T) = \prod_{v=1}^n (y + d_T(v) - 1), \quad w(T) = w_1(T). \quad (17)$$

The root has no weight. Decomposing the unordered collection of root branches and using the labeled exponential formula gives

$$F_y(x) = \sum_{n \geq 0} \frac{x^n}{n!} \sum_{T \in \mathcal{T}_n} w_y(T). \quad (18)$$

Indeed a branch has a root carrying xy and descendants described by F_{y+1} ; the unordered set of branches gives $\exp(xyF_{y+1})$. The same argument with maximum depth at most h gives $A_{1,h}$ at $y = 1$. Consequently

$$n!c_{n,h} = \sum_{\substack{T \in \mathcal{T}_n \\ \text{height}(T)=h}} w(T), \quad a_n = \sum_{h=1}^n n!c_{n,h} \quad (n \geq 1). \quad (19)$$

In particular

$$b_n = \sum_{h=1}^n \frac{c_{n,h}}{n!}. \quad (20)$$

These conclusions can also be proved without combinatorics. At the bottom, $A_{h,h} - 1 = e^{hx} - 1$ has positive coefficients and least degree one. If the successive difference has nonnegative coefficients and least degree j , then the identity $e^U - e^V = e^V(e^{U-V} - 1)$ shows that the difference one level above has nonnegative coefficients and least degree $j + 1$. Thus H_h starts in degree h and the telescoping sum is finite in every coefficient. This proves the formal identity even though F_1 has zero ordinary radius of convergence: a chain alone gives $[x^n]F_1 \geq n!$.

For computational purposes, if $B(x) = \sum B_n x^n / n!$ and $A(x) = \exp(ixB(x)) = \sum A_n x^n / n!$, differentiating A yields the exact integer recurrence

$$A_0 = 1, \quad A_n = \sum_{k=1}^n \binom{n-1}{k-1} k i B_{k-1} A_{n-k}. \quad (21)$$

Starting with $B_0 = 1$ and $B_n = 0$ for $n > 0$, apply this from level h down to 1. Height n already gives the stabilized coefficient a_n . The initial values are

$$1, 1, 5, 61, 1377, 49721, 2625313, 190735749.$$

Lemma 2.1 (Monotonicity). *For $n \geq 1$, $a_n \geq n^2 a_{n-1}$, and hence $b_n \geq b_{n-1}$. In particular a_n is strictly increasing for $n \geq 2$.*

Proof. Write $P_j(y) = j![x^j]F_y$. Restricting the root to one child gives $a_n \geq nP_{n-1}(2)$. In any tree with $n-1$ nonroot vertices, arrange its nonroot depths increasingly as $d_1 \leq \dots \leq d_{n-1}$. Then $d_j \leq j$: a vertex at depth d has ancestors at each depth $1, \dots, d-1$. Thus

$$\prod_{j=1}^{n-1} \frac{d_j + 1}{d_j} \geq \prod_{j=1}^{n-1} \frac{j+1}{j} = n.$$

Apply this to every tree in $P_{n-1}(2)$ to obtain $P_{n-1}(2) \geq nP_{n-1}(1)$. Empty products cover $n=1$. $\square$

3 Critical entrance and sensitivity

Throughout Sections 3–5, put $x_h = 1/(eh)$ and $m = h^{1/3}$. At this real argument let

$$q_i^A = \log A_{i,h}(x_h), \quad q_i^B = \log A_{i,h-1}(x_h),$$

where $q_h^A = 1/e$ and $q_h^B = 0$. Define $u_k^A = 1 - q_{h-k}^A$ and $u_k^B = 1 - q_{h-k}^B$, $0 \leq k \leq h-1$. Both obey the exact recurrence

$$u_k = 1 - (1 - k/h)e^{-u_{k-1}}, \quad u_0^A = \rho, \quad u_0^B = 1. \quad (22)$$

Here $0 < u_k \leq 1$. With $\phi(u) = 1 - e^{-u}$, the terminal states have the important relation $u_0^A = \phi(u_0^B)$. It will fix the sensitivity constant.

Let w solve

$$w''(t) = \frac{1}{2}tw(t), \quad w(0) = 0, \quad w'(0) = 1, \quad (23)$$

and put $U = 2w'/w$. Positive initial slope and $w'' \geq 0$ while $w \geq 0$ imply $w, w' > 0$ for $t > 0$. Direct differentiation and a Taylor series at zero give

$$U' = t - \frac{1}{2}U^2, \quad U(t) = 2/t + \mathcal{O}(t^2) \quad (t \downarrow 0). \quad (24)$$

Lemma 3.1 (Entrance with normalized sensitivity). *Uniformly on each compact interval $0 < d \leq t \leq L < \infty$,*

$$mu_{[mt]}^A \longrightarrow U(t), \quad mu_{[mt]}^B \longrightarrow U(t), \quad (25)$$

$$m^2(u_{[mt]}^B - u_{[mt]}^A) \longrightarrow \frac{2}{w(t)^2}. \quad (26)$$

Proof. For fixed $a \in (0, 1]$, let $v_k = \phi^k(a)$. This sequence decreases to zero and

$$v_{k+1} = v_k - \frac{1}{2}v_k^2 + \mathcal{O}(v_k^3), \quad \frac{1}{v_{k+1}} - \frac{1}{v_k} \longrightarrow \frac{1}{2}.$$

Cesàro summation gives $kv_k \rightarrow 2$. Since ϕ is increasing and 1-Lipschitz on $[0, 1]$, comparison with (22), using the same initial state, gives

$$0 \leq u_k - v_k \leq \sum_{j=1}^k j/h = \frac{k(k+1)}{2h}. \quad (27)$$

For a small fixed $d > 0$, this implies

$$mu_{\lfloor dm \rfloor} = 2/d + \mathcal{O}(d^2) + o_h(1). \quad (28)$$

On $d \leq k/m \leq L$, the scaled values $V_k = mu_k$ are bounded by the same comparison. Taylor expansion of the recurrence is uniform there and gives

$$V_k - V_{k-1} = m^{-1}(k/m - V_{k-1}^2/2) + \mathcal{O}_{d,L}(m^{-2}). \quad (29)$$

For fixed d , discrete Gronwall therefore compares this Euler scheme uniformly on $[d, L]$ to the exact ODE flow from its initial value. Two nonnegative Riccati solutions have difference D satisfying $D' = -(U_1 + U_2)D/2$, so the absolute difference contracts. Compare the flow started in (28) to (24); take $h \rightarrow \infty$ and then $d \downarrow 0$. The discrepancy is $\mathcal{O}(d^2)$ and tends to zero. This proves (25), including uniformity and harmless grid rounding.

For sensitivity use $v_k = \phi^k(1)$ and the autonomous pair v_k, v_{k+1} . Set $\Delta_k = u_k^B - u_k^A$ and $\delta_k = v_k - v_{k+1}$. Their initial differences agree. The exact divided difference

$$J(a, b) = \frac{\phi(b) - \phi(a)}{b - a} = \int_0^1 e^{-(1-r)a - rb} dr \quad (30)$$

has the following property: if both endpoints change by at most E , the logarithm of J changes by at most E . Indeed each integrand changes by a factor between e^{-E} and e^E . Iterating the differences gives

$$\Delta_k = \Delta_0 \prod_{j=1}^k (1 - j/h) J(u_{j-1}^A, u_{j-1}^B), \quad \delta_k = \delta_0 \prod_{j=1}^k J(v_j, v_{j-1}).$$

Use (27) on each endpoint. Uniformly for $k \leq dm$ and large h ,

$$\left| \log \frac{\Delta_k}{\delta_k} \right| \leq C(k^2/h + k^3/h) = \mathcal{O}(d^3) + o_h(1). \quad (31)$$

Since $kv_k \rightarrow 2$, Taylor expansion of $v_k - \phi(v_k)$ gives $k^2\delta_k \rightarrow 2$, so

$$m^2 \Delta_{\lfloor dm \rfloor} = \frac{2}{d^2} \exp\{\mathcal{O}(d^3) + o_h(1)\}. \quad (32)$$

On $d \leq j/m \leq L$, (25) and (30) give

$$\log J(u_{j-1}^A, u_{j-1}^B) = -U(j/m)/m + o_h(1/m)$$

uniformly. Also $\sum_{j \leq Lm} |\log(1 - j/h)| = \mathcal{O}_L(m^2/h) = o(1)$. Hence, for fixed $d \leq t \leq L$,

$$\frac{\Delta_{\lfloor tm \rfloor}}{\Delta_{\lfloor dm \rfloor}} \longrightarrow \exp\left(-\int_d^t U(r) dr\right) = \frac{w(d)^2}{w(t)^2}. \quad (33)$$

Now $w(d) = d + \mathcal{O}(d^4)$. Let $d \downarrow 0$ in (32)–(33) to prove (26). The local logarithmic error used here is $o_h(1/m)$, which is sufficient after summing; an unproved $\mathcal{O}(m^{-2})$ rate of entrance convergence is not needed. $\square$

4 Bulk matching and the endpoint product

Let $T(z)$ be the small real solution of $Te^{-T} = z$, for $0 \leq z \leq 1/e$. In particular $T(0) = 0$ and $T(1/e) = 1$. For $z < 1/e$ define

$$R(z) = \frac{T(z)e^{T(z)}}{(1 - T(z))^2}. \quad (34)$$

We use an approximate solution with its first adiabatic correction; retaining this correction is necessary to recover the amplitude.

Lemma 4.1 (Bulk product matching). *For $i = h - \lfloor tm \rfloor$, write $\varepsilon = 1/(eh)$, $z_j = j\varepsilon$, $T_j = T(z_j)$, and $v_i = 1 - T_i$. For either of the two critical towers,*

$$\limsup_{h \rightarrow \infty} \left| \log \left(v_i \prod_{j=1}^{i-1} \frac{q_j}{T_j} \right) \right| = \mathcal{O}(t^{-3/2}) \quad (t \rightarrow \infty). \quad (35)$$

Proof. Put $Q_j = T_j + \varepsilon R(z_j)$. Differentiation gives

$$\begin{aligned} T' &= e^T/(1 - T), & T'' &= e^{2T}(2 - T)/(1 - T)^3, \\ R' &= e^{2T}(1 + 2T - T^2)/(1 - T)^4, & R &= T(T' + R). \end{aligned}$$

The last identity cancels the first-order residual. Taylor's theorem yields

$$E_j = Q_j - z_j e^{Q_{j+1}}, \quad |E_j| \leq C\varepsilon^2 \frac{T_j}{(1 - T_j)^4} \quad (j < i), \quad (36)$$

provided hw_i^3 exceeds an absolute constant. Here is the relevant uniformity. Writing $v_j = 1 - T_j$, neighboring gaps satisfy $v_j/v_{j+1} \leq 1 + C\varepsilon/v_i^2$. Taylor bounds on the neighboring interval may therefore use v_j . The expansion of Q_{j+1} is $T_j + \varepsilon(T_j' + R_j) + \mathcal{O}(\varepsilon^2 v_j^{-4})$; $z_j e^{T_j} = T_j$, and the remaining exponential quadratic term is also $\mathcal{O}(\varepsilon^2 T_j v_j^{-4})$. This proves (36), including its factor T_j at the left endpoint.

The Airy function and derivative asymptotics, written explicitly in Section 5, imply

$$U(t) = \sqrt{2t} - \frac{1}{2t} + \mathcal{O}(t^{-5/2}). \quad (37)$$

The turning-point expansion of $Te^{-T} = z$ and Lemma 3.1 then give

$$mv_i \longrightarrow \sqrt{2t}, \quad m(q_i - Q_i) = \mathcal{O}(t^{-5/2}) + o_h(1). \quad (38)$$

For fixed sufficiently large t , all Q_j with $j \leq i$ are at most $1 - cv_i$: both T and R increase and $\varepsilon R(z_i)/v_i = \mathcal{O}(t^{-3/2})$. For the true tower use $M = \max(q_i, T_i) < 1$. On $[0, 1]$ the function ue^{-u} increases, so $z_i e^M \leq M$; descending induction gives $q_j \leq M \leq 1 - cv_i$ for $j \leq i$. The same bounds for both arguments in the mean-value theorem show

$$|q_j - Q_j| \leq r|q_{j+1} - Q_{j+1}| + |E_j|, \quad r \leq 1 - c'v_i.$$

For example the derivative is at most $z_i e^{M'}$ with $M' = \max(q_i, Q_i, T_i) \leq 1 - cv_i$; its logarithm is at most $-cv_i$. Summing the geometric propagation therefore gives

$$\sum_{j=1}^i |q_j - Q_j| \leq \frac{C}{v_i} \left(|q_i - Q_i| + \sum_{j=1}^{i-1} |E_j| \right). \quad (39)$$

The monotonicity of the summand in (36), followed by the substitution $dz = e^{-T}(1 - T)dT$, gives

$$\sum_{j < i} |E_j| \leq C(h^{-1}v_i^{-2} + h^{-2}v_i^{-4}).$$

Indeed its integral is bounded by Cv_i^{-2} and the last mesh cell costs $C\varepsilon^2v_i^{-4}$. Equations (38) and (39) now give

$$\limsup_{h \rightarrow \infty} \sum_{j=1}^i |q_j - Q_j| = \mathcal{O}(t^{-3/2}). \quad (40)$$

To convert this additive control to a product without dividing by small T_j , use the exact identity

$$\log(q_j/T_j) = q_{j+1} - T_j.$$

Thus

$$\sum_{j=1}^{i-1} \log(q_j/T_j) = \sum_{j=2}^i (q_j - T_j) + T_i - T_1.$$

Replace $q_j - T_j$ by $\varepsilon R(z_j)$ using (40). The Riemann-sum error is $\mathcal{O}(\varepsilon R(z_i)) = o_h(1)$ for fixed t , and direct substitution gives

$$\int_0^{z_i} R(z) dz = -T_i - \log(1 - T_i). \quad (41)$$

As $T_1 \rightarrow 0$, these formulas prove (35). $\square$

Lemma 4.2 (Endpoint Riemann formula). *If f is absolutely continuous on $[0, 1]$ with $f' \in L^1[0, 1]$, then*

$$\sum_{j=1}^h f(j/h) - h \int_0^1 f(t) dt \longrightarrow \frac{f(1) - f(0)}{2}. \quad (42)$$

Proof. Integrating over each mesh cell makes the difference exactly $\int_0^1 f'(t)\{ht\} dt$, where braces denote fractional part; grid points have measure zero. The bounded functions $\{ht\}$ converge weakly against L^1 functions to $1/2$. Prove this first for interval indicators by summing whole mesh cells, then for step functions, and finally use L^1 approximation and $0 \leq \{ht\} \leq 1$. $\square$

Lemma 4.3 (Critical equilibrium product). *With $T_j = T(j/(eh))$,*

$$\prod_{j=1}^h T_j \sim \sqrt{2\pi eh} e^{-\alpha h}. \quad (43)$$

For fixed $t > 0$ and $i = h - \lfloor tm \rfloor$,

$$\prod_{j=i}^h T_j \longrightarrow \exp\left(-\frac{2\sqrt{2}}{3}t^{3/2}\right). \quad (44)$$

Proof. Since $T_j = z_j e^{T_j}$, the product of z_j is $h!/(eh)^h$ and is treated exactly by Stirling. Apply Lemma 4.2 to $f(t) = T(t/e)$. Its derivative is integrable, as follows either by substituting $t/e = Te^{-T}$ or by its square-root endpoint expansion, so

$$\sum_{j=1}^h T(j/(eh)) = h \int_0^1 T(t/e) dt + \frac{1}{2} + o(1).$$

The identity

$$\int_0^1 \log T(t/e) dt = -\alpha \quad (45)$$

follows by substituting $t = eTe^{-T}$, or from the antiderivative $z \log T(z) + e^{-T(z)}$ of $\log T(z)$. The factorial and this integral give (43); the $1/2$ endpoint term accounts for $\sqrt{e}$. Finally

$$\log T((1-y)/e) = -\sqrt{2y} + \mathcal{O}(y) \quad (y \downarrow 0).$$

Summing for $y = (h-j)/h \leq t/m^2$ gives (44) by a Riemann sum. The total $\mathcal{O}(y)$ remainder is $\mathcal{O}(t^2/m) = o_h(1)$. $\square$

5 Critical finite-height amplitude

The normalized solution of (23) is exactly

$$w(t) = \frac{\pi}{\beta} [\text{Ai}(0) \text{Bi}(\beta t) - \text{Bi}(0) \text{Ai}(\beta t)]. \quad (46)$$

The Airy Wronskian $\text{Ai} \text{Bi}' - \text{Ai}' \text{Bi} = 1/\pi$ verifies both initial conditions. The positive-real function and derivative expansions [6, 7] give

$$w(t) \sim \sqrt{\pi} \text{Ai}(0) \beta^{-5/4} t^{-1/4} e^{(\sqrt{2}/3)t^{3/2}}, \quad (47)$$

$$\frac{2w'(t)}{w(t)} = \sqrt{2t} - \frac{1}{2t} + \mathcal{O}(t^{-5/2}). \quad (48)$$

Equation (48) uses the separate expansion for Bi' as well as for Bi ; it is not obtained by differentiating a bare asymptotic equivalence. This supplies the input (37) in the bulk proof without using a finite-tower conclusion circularly.

Theorem 5.1 (Critical tower equivalent). *As $h \rightarrow \infty$,*

$$H_h(1/(eh)) \sim Kh^{1/6} e^{-\alpha h}, \quad K = \frac{2^{1/6} \sqrt{e}}{\sqrt{\pi} \text{Ai}(0)^2}. \quad (49)$$

Proof. Let $D_i = q_i^A - q_i^B$. Descending the exact q recurrence and applying the mean-value theorem at each level gives

$$D_1 = D_i \prod_{j=1}^{i-1} \tilde{q}_j, \quad q_j^B \leq \tilde{q}_j \leq q_j^A.$$

By squeezing, Lemma 4.1 applies to the product of these intermediate factors. Also $q_1^A, q_1^B \rightarrow 0$: for fixed large t , the bulk bounds keep q_2 bounded and $q_1 = (eh)^{-1} e^{q_2}$. Therefore $H_h/D_1 \rightarrow 1$. Lemma 3.1 gives $D_i \sim 2/(m^2 w(t)^2)$ at $i = h - \lfloor tm \rfloor$. Using the bulk product, (43), (44), and $mv_i \rightarrow \sqrt{2t}$, we find

$$\limsup_{h \rightarrow \infty} \left| \log \frac{H_h(1/(eh)) e^{\alpha h} h^{-1/6}}{2\sqrt{\pi e} t^{-1/2} e^{(2\sqrt{2}/3)t^{3/2}} / w(t)^2} \right| = \mathcal{O}(t^{-3/2}). \quad (50)$$

The power is $h^{1/2} m^{-1} = h^{1/6}$, coming from the equilibrium product, sensitivity, and inverse gap. By (47), the denominator in (50) tends to

$$\frac{2\sqrt{e/\pi} \beta^{5/2}}{\text{Ai}(0)^2} = \frac{2^{1/6} \sqrt{e}}{\sqrt{\pi} \text{Ai}(0)^2} = K.$$

First choose t large so that the logarithmic matching error is small, then let h grow. Sending $t \rightarrow \infty$ proves the limit, rather than only a subsequential relation. $\square$

6 A uniform real critical window

For λ in a fixed compact set $J \subset (-\infty, 1]$, put

$$x_h(\lambda) = \frac{1 + \lambda/m^2}{eh}, \quad m = h^{1/3}. \quad (51)$$

This is positive for all sufficiently large h , uniformly on J . All uniform statements in this section refer to such a fixed compact set.

Theorem 6.1 (Real critical window). *Uniformly for $\lambda \in J$,*

$$H_h(x_h(\lambda)) \sim K(\lambda)h^{1/6}e^{-\alpha h + \alpha \lambda m}, \quad K(\lambda) = \frac{2^{1/6}\sqrt{e}}{\sqrt{\pi} \operatorname{Ai}(-\beta\lambda)^2}. \quad (52)$$

6.1 Parameterized entrance and positive continuation

Define q^A, q^B, u^A, u^B exactly as before but at $x_h(\lambda)$. Their recurrence is now

$$\begin{aligned} u_k &= 1 - (1 + \lambda/m^2)(1 - k/h)e^{-u_{k-1}}, \\ u_0^A &= 1 - (1 + \lambda/m^2)/e, \quad u_0^B = 1. \end{aligned} \quad (53)$$

Let

$$w'_\lambda = \frac{1}{2}(t - \lambda)w_\lambda, \quad w_\lambda(0) = 0, \quad w'_\lambda(0) = 1, \quad U_\lambda = 2w'_\lambda/w_\lambda. \quad (54)$$

We first justify positivity on the whole range being used. For $\lambda \leq 0$, convexity gives $w_\lambda, w'_\lambda > 0$ on $t > 0$. For $0 < \lambda \leq 1$, suppose w'_λ first vanishes at $t \leq \lambda$. Prior to that point, $0 < w_\lambda(r) \leq r$, since $w''_\lambda \leq 0$, and

$$w'_\lambda(t) = 1 - \frac{1}{2} \int_0^t (\lambda - r)w_\lambda(r) dr \geq 1 - \lambda t^2/4 \geq 3/4,$$

a contradiction. Beyond λ , convexity preserves positivity. Therefore U_λ is positive on $t > 0$ and

$$U'_\lambda = t - \lambda - U_\lambda^2/2, \quad U_\lambda(t) = 2/t - (\lambda/3)t + \mathcal{O}_J(t^2). \quad (55)$$

We give the perturbation and continuation details because assuming discrete positivity in advance would be circular. Compare (53) up to $k \leq dm$ with the autonomous pair $\phi^{k+1}(1), \phi^k(1)$. Stop at the first nonpositive discrete state. Before that time the derivative of the recurrence with respect to its state is at most $1 + C_J/m^2$. The initial perturbation is $\mathcal{O}_J(m^{-2})$; the step perturbation is $\mathcal{O}_J(m^{-2} + k/h)$. Discrete Gronwall gives the endpoint error

$$\mathcal{O}_J\left(m^{-2} + k/m^2 + k^2/h\right). \quad (56)$$

Uniformly for starting values in a compact subset of $(0, 1]$, the autonomous iterate is at least $c/(k+1)$; this follows from $k\phi^k(a) \rightarrow 2$, with finitely many initial indices absorbed in c . For $k \leq dm$, the error in (56) is at most $C_J(m^{-2} + (d + d^2)/m)$ whereas the smallest autonomous value is at least $c'/(dm)$. Choose fixed $d > 0$ small enough. This rules out the first crossing, including the initially bounded indices, so (56) holds throughout the entrance segment. It follows that

$$mu_{\lfloor dm \rfloor} = 2/d + \mathcal{O}_J(d + d^2) + o_h(1). \quad (57)$$

For bounded $V = mu$, the recurrence Taylor expansion is

$$V_k - V_{k-1} = m^{-1}(k/m - \lambda - V_{k-1}^2/2) + \mathcal{O}_{d,L,J}(m^{-2}).$$

Fix J and L . The functions U_λ have a common positive lower bound on $(0, L]$, since they diverge uniformly at zero and are positive on compact subintervals. Choose d so that the error in (57) relative to (55) is less than one quarter of that lower bound. The exact Riccati flow from the discrete entrance value cannot reach zero: until a hypothetical first zero, its difference from U_λ contracts. It also stays in a bounded set on $[d, L]$. Standard bounded-interval Euler stability now keeps the discrete trajectory close to that flow. Positivity and boundedness of the discrete trajectory are thus conclusions of a stopped comparison argument. First take $h \rightarrow \infty$, then $d \downarrow 0$. Uniformly on compact positive t intervals and on J ,

$$mu_{[mt]}^A \longrightarrow U_\lambda(t), \quad mu_{[mt]}^B \longrightarrow U_\lambda(t). \quad (58)$$

For the sensitivity, use the same integral divided difference (30). Summing (56) over $k \leq dm$ changes its logarithmic product by $\mathcal{O}_J(d^2 + d^3) + o_h(1)$. The additional recurrence factor $1 + \lambda/m^2$ contributes only $\mathcal{O}_J(k/m^2) = o_h(1)$, and the initial difference ratio is $1 + \lambda/m^2$. The autonomous normalization is still 2. Hence

$$m^2 \Delta_{[dm]} = \frac{2}{d^2} \exp\{\mathcal{O}_J(d^2 + d^3) + o_h(1)\}.$$

Propagate from d to t using (58), with the uniform $o_h(1/m)$ logarithmic error as before, and let $d \downarrow 0$ to obtain

$$m^2(u_{[mt]}^B - u_{[mt]}^A) \longrightarrow \frac{2}{w_\lambda(t)^2}. \quad (59)$$

6.2 Uniform bulk matching

Set $\varepsilon = x_h(\lambda)$, $z_j = j\varepsilon$, $i = h - \lfloor tm \rfloor$, and $T_j = T(z_j)$ only for $j \leq i$. Choose a fixed $t > 1 + \max(0, \sup J)$, so $z_i < 1/e$ for large h . The same approximation $Q_j = T_j + \varepsilon R(z_j)$ has the uniform residual

$$|Q_j - z_j e^{Q_{j+1}}| \leq C_J \varepsilon^2 T_j / (1 - T_j)^4.$$

Writing $v_i = 1 - T_i$, the entrance estimates give

$$mv_i \longrightarrow \sqrt{2(t - \lambda)}, \quad (60)$$

$$m(q_i - Q_i) \longrightarrow \sqrt{2(t - \lambda)} - U_\lambda(t) - \frac{1}{2(t - \lambda)}. \quad (61)$$

The normalized Airy representation is

$$w_\lambda(t) = \frac{\pi}{\beta} [\text{Ai}(-\beta\lambda) \text{Bi}(\beta(t - \lambda)) - \text{Bi}(-\beta\lambda) \text{Ai}(\beta(t - \lambda))]. \quad (62)$$

The Wronskian checks its initial conditions. Its coefficient of the growing Bi term is positive: a negative coefficient contradicts $w_\lambda > 0$ at large t , and zero would make a multiple of the decaying Ai, contradicting $w_\lambda, w'_\lambda > 0$. In particular $\text{Ai}(-\beta\lambda) > 0$ for $\lambda \leq 1$, bounded away from zero on J . The positive-real function and derivative expansions give, uniformly on J ,

$$w_\lambda(t) \sim \sqrt{\pi} \text{Ai}(-\beta\lambda) \beta^{-5/4} (t - \lambda)^{-1/4} e^{(\sqrt{2}/3)(t - \lambda)^{3/2}}, \quad (63)$$

$$U_\lambda(t) = \sqrt{2(t - \lambda)} - \frac{1}{2(t - \lambda)} + \mathcal{O}_J((t - \lambda)^{-5/2}). \quad (64)$$

Thus (61) is $\mathcal{O}_J((t - \lambda)^{-5/2})$. Every residual, neighbor-gap, scalar contraction, and residual-sum estimate in the proof of Lemma 4.1 applies on $[0, z_i]$, with uniform constants on J . More explicitly,

$v_i \asymp m^{-1}\sqrt{t-\lambda}$; the summed residual after division by v_i is $\mathcal{O}_J((t-\lambda)^{-3/2}) + o_h(1)$ and the boundary error after that division is $\mathcal{O}_J((t-\lambda)^{-3}) + o_h(1)$. The exact product identity and (41) are unchanged. We obtain

$$v_i \prod_{j=1}^{i-1} (q_j/T_j) = \exp\{\mathcal{O}_J(t^{-3/2}) + o_h(1)\} \quad (65)$$

in the same iterated-limit sense as (35). Squeezing also proves this for the intermediate divided-difference factors. No value of the real branch T beyond z_i is used.

6.3 A truncated equilibrium product

For $\lambda > 0$ the real turning point occurs before level h , so we must not extend the full product of Lemma 4.3 past that point. Put

$$\mathcal{H} = \frac{h}{1 + \lambda/m^2} = h - \lambda m + \mathcal{O}_J(m^{-1}), \quad r = i - 1, \quad b = r/\mathcal{H}. \quad (66)$$

Then $z_j = j/(e\mathcal{H})$ and $1 - b = (t - \lambda)/m^2 + \mathcal{O}_J(m^{-3})$. Factoring $T(z) = ze^{T(z)}$ and applying Stirling gives

$$\log \prod_{j=1}^r T(j/(e\mathcal{H})) = \mathcal{H} \int_0^b \log T(v/e) dv + \frac{1}{2} \log(2\pi r) + \frac{1}{2} T(b/e) + o_h(1). \quad (67)$$

Here the singular $\log(v/e)$ part is handled by the factorial, never by an endpoint derivative estimate. For completeness, the error for the remaining function $f(v) = T(v/e)$ is $\int_0^b f'(v) \{\mathcal{H}v\} dv$. Extend $f' \mathbf{1}_{[0,b]}$ by zero to $[0, 1]$. These truncated derivatives tend to f' in L^1 , uniformly on J ; the bounded fractional parts have the same weak limit $1/2$ for real $\mathcal{H} \rightarrow \infty$. This proves the $o_h(1)$ in (67), including its moving endpoint.

By (45) and the square-root expansion at 1,

$$\mathcal{H} \int_0^b \log T(v/e) dv = -\alpha \mathcal{H} + \frac{2\sqrt{2}}{3} (t - \lambda)^{3/2} + o_h(1).$$

The $\mathcal{O}(m^{-3})$ floor error costs $o_h(1)$ in this exponent. As $r/h \rightarrow 1$ and $T(b/e) \rightarrow 1$, we have proved

$$\prod_{j=1}^{i-1} T(j\varepsilon) \sim \sqrt{2\pi e h} e^{-\alpha h + \alpha \lambda m + (2\sqrt{2}/3)(t-\lambda)^{3/2}}, \quad (68)$$

uniformly on J for fixed sufficiently large t .

6.4 Completion of the window theorem

The exact difference product still gives $H_h \sim D_i \prod_{j < i} \tilde{q}_j$, because $q_1^A, q_1^B \rightarrow 0$ uniformly on J . Combine (59), (60), (65), and (68). The ratio of $H_h e^{\alpha h - \alpha \lambda m} h^{-1/6}$ to

$$\frac{2\sqrt{\pi e} (t - \lambda)^{-1/2} e^{(2\sqrt{2}/3)(t-\lambda)^{3/2}}}{w_\lambda(t)^2}$$

has limiting logarithmic error $\mathcal{O}_J(t^{-3/2})$. By (63) this last display tends uniformly on J to $K(\lambda)$. Take $h \rightarrow \infty$ first and then $t \rightarrow \infty$. This proves Theorem 6.1 with the asserted compact uniformity.

7 The Airy size law and its density

Define N_h by (10). Since H_h is entire, positive at $x_h = 1/(eh)$, and has nonnegative coefficients, this is a probability law with support in the integers $n \geq h$. Set

$$X_h = (N_h - \alpha h)/m^2.$$

For real λ near zero,

$$M_h(\lambda) = \mathbb{E}e^{\lambda X_h} = e^{-\alpha \lambda m} \frac{H_h(x_h e^{\lambda/m^2})}{H_h(x_h)}. \quad (69)$$

The effective parameter in Theorem 6.1 is

$$\lambda_{\text{eff}} = m^2(e^{\lambda/m^2} - 1) = \lambda + \mathcal{O}_J(m^{-2}).$$

Even in the exponential factor the discrepancy vanishes: $\alpha m(\lambda_{\text{eff}} - \lambda) = \mathcal{O}_J(m^{-1})$. Uniformity of the window theorem therefore gives

$$M_h(\lambda) \longrightarrow M(\lambda) = \frac{\text{Ai}(0)^2}{\text{Ai}(-\beta\lambda)^2} \quad (70)$$

on an open real interval containing zero. The moment-generating-function continuity theorem implies weak convergence $X_h \xrightarrow{d} X$.

7.1 Analytic continuation to every imaginary argument

One still needs to identify the characteristic function for every real Fourier argument, not just near zero. Choose real $a < 0 < b < 1$ inside the interval used in (70). The complex MGFs are entire and, for $a \leq \Re z \leq b$,

$$|M_h(z)| \leq M_h(\Re z) \leq M_h(a) + M_h(b). \quad (71)$$

The right-hand side is uniformly bounded. Vitali's theorem consequently gives local analytic convergence throughout the vertical strip to the analytic continuation of (70). To spell out the uniqueness step, normal-family compactness supplies a convergent subsequence on each compact subset; every such subsequential limit agrees on the real interval and hence, by the identity theorem, on the entire connected strip. Thus all subsequences have the same limit. The Airy function has only negative real zeros [8], and the positivity proved after (62) excludes a zero of $\text{Ai}(-\beta z)$ on the chosen real interval. The displayed candidate is therefore holomorphic on the strip. In particular

$$\mathbb{E}e^{itX} = \psi(t) = \frac{\text{Ai}(0)^2}{\text{Ai}(-\beta it)^2} \quad (t \in \mathbb{R}). \quad (72)$$

The complex Airy asymptotic on the rays with arguments $\pm\pi/2$ [7] gives

$$|\text{Ai}(-\beta it)^{-2}| = \mathcal{O}\left(|t|^{1/2} e^{-(2/3)|t|^{3/2}}\right) \quad (|t| \rightarrow \infty). \quad (73)$$

Thus $\psi \in L^1(\mathbb{R})$, and Fourier inversion gives the bounded continuous density

$$p(v) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itv} \psi(t) dt. \quad (74)$$

7.2 The exact density at zero

The Wronskian and the connection identity [6] imply

$$\left(\frac{\text{Bi}}{\text{Ai}}\right)'(z) = \frac{1}{\pi \text{Ai}(z)^2}, \quad (75)$$

$$\text{Bi}(z) = i \text{Ai}(z) + 2e^{-i\pi/6} \text{Ai}(ze^{-2\pi i/3}). \quad (76)$$

At $z = iR$, the rotated Airy argument in (76) has angle $-\pi/6$. Its Airy function decays exponentially, while $\text{Ai}(iR)$ grows exponentially. Hence

$$\frac{\text{Bi}(iR)}{\text{Ai}(iR)} \longrightarrow i.$$

Complex conjugation gives the limit $-i$ at $z = -iR$. There are no Airy zeros on the imaginary axis. Integrating (75) upward along it gives

$$\int_{-i\infty}^{i\infty} \frac{dz}{\text{Ai}(z)^2} = \pi \left[\frac{\text{Bi}}{\text{Ai}} \right]_{-i\infty}^{i\infty} = 2\pi i. \quad (77)$$

The integral is absolutely convergent by (73). Under $z = -\beta it$, the increasing real t axis maps to the downward imaginary axis. Since $dt = i dz/\beta$, reversing the orientation in (77) proves

$$p(0) = \frac{\text{Ai}(0)^2}{2\pi} \int_{\mathbb{R}} \frac{dt}{\text{Ai}(-\beta it)^2} = \frac{\text{Ai}(0)^2}{\beta} > 0. \quad (78)$$

This calculation fixes the normalization exactly. It does not yet imply anything about the individual lattice probabilities $\mathbb{P}(N_h = n)$.

8 Fourier minor arcs from positive factorization

Let

$$\varphi_h(\theta) = \mathbb{E} e^{i\theta N_h} = \frac{H_h(x_h e^{i\theta})}{H_h(x_h)}, \quad x_h = 1/(eh). \quad (79)$$

The argument in this section does not use the critical equivalent or the Airy window theorem.

Theorem 8.1 (Uniform minor-arc bound). *There exist constants $c > 0$, $A > 0$, and h_0 such that, for $h \geq h_0$,*

$$|\varphi_h(\theta)| \leq \begin{cases} e^{-ch|\theta|^{3/2}}, & Ah^{-2/3} \leq |\theta| \leq 1/8, \\ e^{-ch}, & 1/8 \leq |\theta| \leq \pi. \end{cases} \quad (80)$$

8.1 An exact positive factorization

For a complex variable z , abbreviate $A_i(z) = A_{i,h}(z)$, $B_i(z) = A_{i,h-1}(z)$, $\Delta_i(z) = A_i(z) - B_i(z)$, where $B_h = 1$. Define

$$D_i(z) = \int_0^1 \exp(izt\Delta_{i+1}(z)) dt \quad (1 \leq i < h).$$

Here i is the integer level, not the imaginary unit i . The exact difference identity is

$$\Delta_i(z) = izB_i(z)\Delta_{i+1}(z)D_i(z), \quad \Delta_h(z) = e^{hz} - 1.$$

Iteration gives an identity of entire functions,

$$H_h(z) = (e^{hz} - 1)(h-1)!z^{h-1} \prod_{i=1}^{h-1} B_i(z)D_i(z). \quad (81)$$

Each factor has nonnegative Taylor coefficients. For D_i this follows from the nonnegative coefficients and zero constant term of Δ_{i+1} , followed by coefficientwise integration over $0 \leq t \leq 1$. Normalize each factor by its positive value at x_h and apply the triangle inequality separately. The monomial has normalized modulus one, and all other normalized moduli are at most one. Dropping the terminal and D_i factors gives

$$|\varphi_h(\theta)| \leq \prod_{i=1}^{h-1} \frac{|B_i(x_h e^{i\theta})|}{B_i(x_h)}. \quad (82)$$

This remains true if a dropped factor vanishes on the circle.

8.2 Homogeneous tree comparison

Write $Q_i(z) = izB_{i+1}(z)$, the globally defined entire exponent of B_i ; no choice of logarithm is involved. All its coefficients are nonnegative. Define

$$T_0(z) = 0, \quad T_{j+1}(z) = ze^{T_j(z)}, \quad T(z) = \sum_{r \geq 1} \frac{r^{r-1}}{r!} z^r. \quad (83)$$

The final series follows from Lagrange inversion and is the same small solution $Te^{-T} = z$ used earlier. Coefficientwise,

$$Q_i(z) \succeq T_{h-i}(iz), \quad T_L(z) \preceq T(z). \quad (84)$$

The first comparison starts at $i = h-1$ with equality $Q_{h-1}(z) = (h-1)z = T_1((h-1)z)$. Descending induction uses that every deeper level parameter is at least i and exponentiation preserves nonnegative coefficient inequalities. Put $z_i = ix_h = i/(eh)$ and

$$\mathcal{D}_L(z, \theta) = \sum_{r \geq 1} [z^r] T_L(z) z^r (1 - \cos(r\theta)). \quad (85)$$

In (85) the coefficient notation refers to the power series before evaluating its argument; equivalently, if $T_L(z) = \sum t_{L,r} z^r$, then $\mathcal{D}_L = \sum t_{L,r} z^r (1 - \cos(r\theta))$. Since $1 - \cos(r\theta) \geq 0$, applying (84) to the real-part loss of each exponent in (82) is legitimate and yields

$$|\varphi_h(\theta)| \leq \exp \left(- \sum_{i=1}^{h-1} \mathcal{D}_{h-i}(z_i, \theta) \right). \quad (86)$$

The comparison is on nonnegative coefficient sums, not on oscillatory complex values.

8.3 A finite-iteration tail bound

For $0 \leq z < 1/e$, set $\tau = T(z)$, $\epsilon = 1 - \tau$, and $d_j = \tau - T_j(z)$. Then

$$d_{j+1} = \tau(1 - e^{-d_j}), \quad 0 \leq d_0 = \tau \leq 1.$$

For $0 \leq u \leq 1$, Taylor's inequality gives $1 - e^{-u} \leq u - u^2/3$. Consequently

$$d_{j+1} \leq d_j - d_j^2/3, \quad d_{j+1} \leq \tau d_j.$$

The first inequality gives $d_j \leq 3/(j+3)$: its reciprocal increases by at least $1/3$, starting from $1/d_0 \geq 1$ (the zero case is immediate). Apply the second inequality after $j = \lfloor L/2 \rfloor$ to obtain

$$d_L \leq \frac{6}{L} e^{-\epsilon L/2}, \quad d_L \leq 6\epsilon e^{-\epsilon L/2} \quad \text{if } \epsilon L \geq 1. \quad (87)$$

Positivity of $T - T_L$ and $0 \leq 1 - \cos \leq 2$ imply

$$\mathcal{D}_L(z, \theta) \geq \mathcal{D}_\infty(z, \theta) - 2d_L, \quad \mathcal{D}_\infty(z, \theta) = \sum_{r \geq 1} \frac{r^{r-1} z^r}{r!} (1 - \cos(r\theta)). \quad (88)$$

This uses only real subcritical contraction, with no assumed complex fixed-point estimate.

8.4 A block of near-terminal levels

By symmetry suppose $0 < \theta \leq 1/8$. Select integer levels with

$$k = h - i \in [h\theta, 2h\theta], \quad L = k, \quad \delta = k/h, \quad z_i = (1 - \delta)/e.$$

Thus $\theta \leq \delta \leq 2\theta \leq 1/4$. For integers $r \in [1/\theta, 2/\theta]$ we have $r\theta \in [1, 2]$, so $1 - \cos(r\theta) \geq 1 - \cos 1$. The elementary Stirling bound $r! \leq 3\sqrt{r}(r/e)^r$, valid for $r \geq 1$, gives

$$\frac{r^{r-1} z_i^r}{r!} \geq \frac{(1 - \delta)^r}{3r^{3/2}} \geq \frac{e^{-8}}{3r^{3/2}}.$$

Indeed $\log(1 - \delta) \geq -2\delta$ and $r\delta \leq 4$. The degree interval contains at least $7/(8\theta)$ integers, each with $r^{-3/2} \geq (\theta/2)^{3/2}$. Hence

$$\mathcal{D}_\infty(z_i, \theta) \geq c_0 \sqrt{\theta}, \quad c_0 = \frac{7(1 - \cos 1)e^{-8}}{24 \cdot 2^{3/2}} > 0. \quad (89)$$

Write $\epsilon_i = 1 - T(z_i)$. The exact gap identity is

$$\delta = 1 - (1 - \epsilon_i)e^{\epsilon_i} = \int_0^{\epsilon_i} v e^v dv.$$

As $0 \leq \epsilon_i \leq 1$, it follows that

$$\sqrt{2/e} \sqrt{\theta} \leq \epsilon_i \leq 2\sqrt{\theta}, \quad \epsilon_i L \geq \sqrt{2/e} h \theta^{3/2}. \quad (90)$$

Put $X = h\theta^{3/2}$. For a fixed sufficiently large M and $X \geq M$, (87) and (90) yield

$$2d_L \leq 24\sqrt{\theta} e^{-\sqrt{2/e} X/2} \leq (c_0/2) \sqrt{\theta}.$$

Here choose M so that $\sqrt{2/e} M \geq 1$ and $24e^{-\sqrt{2/e} M/2} \leq c_0/2$. Equations (88) and (89) then give $\mathcal{D}_L(z_i, \theta) \geq (c_0/2) \sqrt{\theta}$ for every selected level. Their number is at least $h\theta - 1 \geq h\theta/2$ after increasing M . All indices lie in $1 \leq k \leq h - 1$ for large h , because $k \leq h/4$ and the lower cutoff makes $h\theta$ large. Substitution in (86) gives

$$|\varphi_h(\theta)| \leq e^{-(c_0/4)h\theta^{3/2}} \quad \text{if } 0 < \theta \leq 1/8, \quad \theta \geq M^{2/3} h^{-2/3}. \quad (91)$$

8.5 Large arcs and lattice resonances

For $1/8 \leq |\theta| \leq \pi$, keep only the degree-one coefficient $Q_i(z) \succeq iz$. At every level $i \in [h/4, h/2]$ its real-part loss is at least

$$ix_h(1 - \cos \theta) \geq (4e)^{-1}(1 - \cos(1/8)) > 0.$$

There are a positive constant times h such levels. Equation (82) gives $|\varphi_h(\theta)| \leq e^{-ch}$ uniformly. Together with (91), this proves Theorem 8.1. In particular no lattice resonance away from zero is left untreated.

9 Uniform lattice transfer

We now combine the pointwise characteristic limit of Section 7 with the separately proved Fourier bounds. For integration define

$$\Phi_h(t) = e^{-it\alpha h/h^{2/3}} \varphi_h(t/h^{2/3})$$

on $|t| \leq \pi h^{2/3}$ and extend it by zero outside that interval. For each fixed t , $\Phi_h(t) \rightarrow \psi(t)$. On every fixed compact interval, dominated convergence applies because $|\Phi_h| \leq 1$. On $A \leq |t| \leq h^{2/3}/8$, Theorem 8.1 gives $|\Phi_h(t)| \leq e^{-c|t|^{3/2}}$. On the remaining large arc the integral is at most $2\pi h^{2/3} e^{-ch} \rightarrow 0$. Since ψ is integrable,

$$\int_{\mathbb{R}} |\Phi_h(t) - \psi(t)| dt \rightarrow 0. \quad (92)$$

The lattice inversion formula is

$$h^{2/3} \mathbb{P}(N_h = n) = \frac{1}{2\pi} \int_{-\pi h^{2/3}}^{\pi h^{2/3}} e^{-it(n-\alpha h)/h^{2/3}} \Phi_h(t) dt.$$

Compare this with (74); the absolute error, uniformly in every integer n , is at most $(2\pi)^{-1}$ times (92). We have proved the uniform local limit (12). In particular, for some absolute C and every integer n and $h \geq 1$,

$$\mathbb{P}(N_h = n) \leq Ch^{-2/3}. \quad (93)$$

For sufficiently large h this follows from the uniform local limit and boundedness of p ; enlarge C over finitely many smaller heights. It can also be obtained directly by integrating Theorem 8.1, bounding the central arc by its width.

10 All-height domination and the coefficient sum

The exact probability definition gives

$$\frac{c_{n,h}}{n!} = \frac{(eh)^n}{n!} H_h(1/(eh)) \mathbb{P}(N_h = n). \quad (94)$$

Use the critical equivalent and (93), increasing a constant for the finitely many small h , to get the global bound

$$\frac{c_{n,h}}{n!} \leq C \frac{e^n}{n!} h^{n-1/2} e^{-\alpha h} \quad (h \geq 1, n \geq 0). \quad (95)$$

The left side vanishes when $h > n$. This bound is available at every height; it does not extend a compact-window asymptotic into the tails.

Let

$$\mathcal{W}_n = \{h \in \mathbb{Z} : |h - n/\alpha| \leq n^{3/5}\}. \quad (96)$$

On this window $h \sim n/\alpha$ and $(n - \alpha h)/h^{2/3} \rightarrow 0$ uniformly. Continuity and positivity of $p(0)$, the uniform local limit, and the critical equivalent yield

$$\frac{c_{n,h}}{n!} = (Kp(0) + o(1)) \frac{e^n}{n!} h^{n-1/2} e^{-\alpha h} \quad \text{uniformly for } h \in \mathcal{W}_n. \quad (97)$$

The window is wider than $\sqrt{n}$ and narrower than $n^{2/3}$; these two inequalities are the reason for the convenient exponent $3/5$. As $\alpha > 1$, all its heights lie in $1 \leq h \leq n$ for large n .

Lemma 10.1 (Discrete Laplace sum with tails). *For $f_n(t) = t^{n-1/2} e^{-\alpha t}$ on $t > 0$,*

$$\sum_{h \geq 1} f_n(h) \sim \int_0^\infty f_n(t) dt = \frac{\Gamma(n + 1/2)}{\alpha^{n+1/2}}. \quad (98)$$

Restricting the sum to $\mathcal{W}_n$ preserves this equivalent.

Proof. A nonnegative unimodal function has sum-integral error at most twice its maximum, by applying the left and right rectangle comparisons on its two monotone pieces. Here the maximum is attained at $t = (n - 1/2)/\alpha$. Stirling gives

$$\frac{\int_0^\infty f_n(t) dt}{\max_{t>0} f_n(t)} \sim \frac{\sqrt{2\pi n}}{\alpha}.$$

Thus the relative error is $\mathcal{O}(n^{-1/2})$. For the tails, write $t = (n/\alpha)(1 + u)$. Relative to the central value,

$$\log \frac{f_n(t)}{f_n(n/\alpha)} = n(\log(1 + u) - u) - \frac{1}{2} \log(1 + u). \quad (99)$$

On $|u| \leq 1/2$, the first term is at most $-cnu^2$. On either side outside this fixed central region the exponent has a strictly negative gap, and toward $u = \infty$ it is eventually bounded above by $-cnu$ after adjusting constants; toward $u = -1$ it tends to $-\infty$. The boundaries of $\mathcal{W}_n$ have $|u| = \alpha n^{-2/5}$, so their exponential loss is at least $cn^{1/5}$. Integrating these elementary bounds shows that the omitted integrals are at most a polynomial in n times $e^{-cn^{1/5}}$ times the full integral. Apply the rectangle comparison separately to each monotone tail: its sum differs from its integral by at most twice its boundary maximum, which satisfies the same bound. This proves that the omitted discrete mass is negligible. $\square$

We can now prove Theorem 1.1. Sum (97) on $\mathcal{W}_n$ and use (95) with Lemma 10.1 outside it. The latter contribution is negligible, so

$$\begin{aligned} b_n &\sim Kp(0) \frac{e^n}{n!} \frac{\Gamma(n + 1/2)}{\alpha^{n+1/2}} \\ &\sim \frac{Kp(0)}{\sqrt{\alpha}} \rho^{-n} n^{-1/2}. \end{aligned} \quad (100)$$

By (1) and (78),

$$\frac{Kp(0)}{\sqrt{\alpha}} = \frac{2^{1/6} \sqrt{e}}{\sqrt{\pi} \beta \sqrt{\alpha}} = \sqrt{\frac{2}{\pi \rho}}. \quad (101)$$

The two factors $\text{Ai}(0)^2$ cancel only after the lattice extraction and height summation have both been justified. Multiplying by $(n!)^2 \sim 2\pi n(n/e)^{2n}$ proves (5), and taking logarithms proves (6). This completes the coefficient theorem.

11 Gaussian height fluctuations

For the weighted-tree law of Theorem 1.2,

$$\mathbb{P}(L_n = h) = \frac{n!c_{n,h}}{a_n} = \frac{c_{n,h}/n!}{b_n}. \quad (102)$$

Equations (97), (100), and Lemma 10.1 show that throughout $\mathcal{W}_n$ this law is uniformly asymptotic to the normalized positive kernel

$$\frac{h^{n-1/2}e^{-\alpha h}}{\sum_{j \geq 1} j^{n-1/2}e^{-\alpha j}}.$$

Equation (95) and the tail part of Lemma 10.1 show that both distributions give asymptotically vanishing mass to the complement of $\mathcal{W}_n$. It follows in particular that their total variation distance tends to zero.

For fixed bounded s , set $h = n/\alpha + s\sqrt{n}/\alpha$, with grid spacing $\alpha/\sqrt{n}$ in the variable s . Taylor expansion of (99) at $u = s/\sqrt{n}$ gives, uniformly on bounded s intervals,

$$\log \frac{f_n(h)}{f_n(n/\alpha)} = -s^2/2 + o(1).$$

The same quadratic bound used in Lemma 10.1 gives uniformly small normalized tails as $|s|$ becomes large, with the farther tails exponentially small. Riemann sums and the normalizing integral therefore imply convergence of this kernel to the standard normal density in the s coordinate. Total variation comparison transfers the weak limit to L_n , proving (13).

This statement centers at n/α and scales by $\sqrt{n}/\alpha$. We do not identify these with exact finite- n mean and standard deviation, or infer moment expansions merely from weak convergence. Nor does this argument say that the Airy law itself becomes Gaussian: that law describes size under a fixed-height Boltzmann measure, whereas this one describes height under the fixed-size tree measure.

12 Refined inversion and the integer threshold

Put

$$A = 2\sqrt{2\pi/\rho}, \quad f(t) = 2t \log \left(\frac{t}{e\sqrt{\rho}} \right). \quad (103)$$

Because $2 + \log \rho = 1 + \log(e - 1)$, Theorem 1.1 gives

$$\log a_n = f(n) + \frac{1}{2} \log n + \log A + o(1). \quad (104)$$

Let g be the continuous piecewise-linear interpolation through these integer log values on the eventually increasing portion. The second derivative of $f(t) + \frac{1}{2} \log t$ is $\mathcal{O}(1/t)$. Linear interpolation therefore changes the smooth expression by $\mathcal{O}(1/t)$, and the integer $o(1)$ error remains uniformly $o(1)$ after interpolation. Thus

$$g(t) = f(t) + \frac{1}{2} \log t + \log A + o(1) \quad (t \rightarrow \infty, t \in \mathbb{R}). \quad (105)$$

The positive real solution of $f(x) = L = \log Y$ is

$$x = \frac{L}{2W_0(L/(2e\sqrt{\rho}))}.$$

To verify this, set $w = \log(x/(e\sqrt{\rho}))$; then $we^w = L/(2e\sqrt{\rho})$ and $x = L/(2w)$. Also

$$f'(x) = 2\log(x/\sqrt{\rho}), \quad f''(x) = 2/x. \quad (106)$$

Let $z(Y) = g^{-1}(L)$. First $z \sim x$ follows from (105) and the leading term $2t \log t$. On the interval between x and z , the derivative in (106) is comparable to $\log x$. The correction in (105) is $\mathcal{O}(\log z)$, so the mean-value theorem improves this to $z - x = \mathcal{O}(1)$. Taylor expansion with that bounded displacement now yields

$$\begin{aligned} z &= x - \frac{\frac{1}{2} \log x + \log A}{2 \log(x/\sqrt{\rho})} + o(1/\log x) \\ &= x - \frac{1}{4} - \frac{\log(8\pi/\sqrt{\rho})}{4 \log(x/\sqrt{\rho})} + o(1/\log x). \end{aligned} \quad (107)$$

For the algebra in the second line use $A^2 = 8\pi/\rho$.

By Lemma 2.1, for all sufficiently large thresholds the integer inverse is exactly

$$N(Y) = \lceil z(Y) \rceil. \quad (108)$$

Equation (107) implies that some positive $\epsilon(Y) \rightarrow 0$ bounds $|z(Y) - r(Y)| \log x$ from above. Monotonicity of the ceiling proves (16), completing Theorem 1.3. The envelope is not explicitly estimated by the proof. Thresholds may approach an integer crossing arbitrarily closely, so replacing this two-sided sandwich by one ceiling of $r(Y)$ would not be justified.

[write] **7 October 2026.** In the transseries volume’s terms (front matter, “The inverses and the transseries volume”), (108) is an instance of item (1) of `p0:thm:staircase`, with the piecewise-linear g as admissible interpolation, and the sandwich follows from it as item (2) does; the Lambert base x is an instance of `p0:prop:factorial-core`. Part II carries $r(Y)$ one term further ((222)).

13 Exact verification and reproducibility

The accompanying archive contains this TeX source and PDF, authored Python code, small attributed integer fixtures, generated verification receipts, and a deterministic offline builder. The core arithmetic and package checks use only the Python standard library; optional arbitrary-precision Airy diagnostics are isolated from the exact verification.

13.1 Three finite descriptions of the coefficients

The primary coefficient computation uses (21) with exact integers. It recomputes all coefficients through $n = 48$ and the exact-height decomposition, rather than importing a floating approximation. A triangular truncation of the same defining recurrence is a useful second implementation but is not represented as an independent combinatorial model.

The first independent combinatorial check enumerates all Prüfer strings on $n + 1$ labels, roots each resulting labeled tree at 0, computes all depths, and accumulates their product by maximum depth. For $n = 1, \dots, 7$ the total number of trees checked is

$$\sum_{n=1}^7 (n+1)^{n-1} = 280392.$$

It verifies every one of the 28 positive-height cells in that range against (19). The one-vertex root object is checked separately.

A further exact method sums by level populations. If the positive level sizes are $n_1, \dots, n_h$ with sum n , then the labels can be assigned in $n! / \prod_j n_j!$ ways, and each vertex at level $j \geq 2$ chooses one of the n_{j-1} parents at the preceding level. For shift y , the total contribution of this profile is

$$\frac{n!}{\prod_{j=1}^h n_j!} \prod_{j=2}^h n_{j-1}^{n_j} \prod_{j=1}^h (y + j - 1)^{n_j}. \quad (109)$$

Every such assignment is a rooted tree and has exactly the prescribed levels, so summing (109) over positive compositions gives another independent finite formula. The package checks all height cells through $n = 12$ using this formula. These checks confirm the weighting and normalization, not the limiting constants by numerical extrapolation.

13.2 Provenance and deliberately bounded claims

The official displayed A096537 entry supplies 15 terms, indices 0 through 14, which are checked exactly [1]. Its linked table advertises indices through 200, but that file was not retrieved or used in this package. The coefficients from 15 through 48 in the regression fixture are explicitly labeled generated data; they are not additional official OEIS comparisons. Nine complete polynomial rows of A096542, through $n = 8$, are also checked at the rational shifts $1/2, 1, 2, 3$ [2]. The defining series and these rows imply $T(n, 1) = n a_{n-1}$, in agreement with the cross-reference on A096537; the retrieved A096542 display $T(n, 1) = n a_n$ has an index mismatch. That display is not used as an input identity.

Remark 13.1 ([write] The A096542 formula for $T(n, 1)$ is off by one index). [Added 7 October 2026, batch 110.] OEIS A096542 (revision #11, 19 December 2017, live on 7 October 2026), whose rows are $T(n, k) = [y^k] P_n(y)$, has the formula line “ $T(n, 1) = n * A096537(n)$ ”. The correct identity is

$$T(n, 1) = n a_{n-1} \quad (n \geq 1).$$

Proof: differentiating $F_y = \exp(xy F_{y+1})$ in y at $y = 0$ gives $[y] F_y = x F_1$, so $T(n, 1)/n! = [x^n] x F_1(x) = a_{n-1}/(n-1)!$ (Part II proves the same in Section 28). The entry’s line is false for every $n \geq 2$, because $a_n \geq n^2 a_{n-1} > a_{n-1}$ there (Lemma 2.1). The entry’s own rows show it: row 2 is 0, 2, 3, so $T(2, 1) = 2 = 2a_1$, while $2a_2 = 10$; likewise $T(3, 1) = 15 = 3 \cdot 5$, $T(4, 1) = 244 = 4 \cdot 61$ and $T(5, 1) = 6885 = 5 \cdot 1377$, against $3 \cdot 61$, $4 \cdot 1377$ and $5 \cdot 49721$. A096537’s own line “ $a(n) = A096542(n+1, 1)/(n+1)$ ” is the correct identity. No correction has been submitted to the OEIS.

These finite checks include coefficient nonnegativity, the all-path diagonal $n! c_{n,n} = (n!)^2$, and domain/fixture rejection tests. Supporting shifted or height-marked finite checks do not assert a shifted-family prefactor theorem or an all-orders asymptotic extension of this report.

13.3 Running and rebuilding

From the extracted package, the mandatory replay is

```
python3 -I -S -B reproduce.py
```

It runs the exact checker both normally and under `-O` and requires byte-identical canonical JSON. Mathematical validity checks use explicit exceptions rather than removable Python assertions. The guard suite is `test_build.py`; the extracted release inventory and its hashes are checked by `verify_manifest.py`.

For the complete PDF and ZIP build, use `build.py -output` with a fresh external directory whose parent already exists. The builder rejects existing output paths, source/output overlap, traversal components, symlinks, wrong types, unexpected inventory, changed sources, and malformed receipts. It runs the guard suite and exact replay in both Python modes, then invokes installed pdfLaTeX in an isolated environment for three passes with shell escape disabled. The settled final log must have no warnings, unresolved references, missing glyphs, or overfull/underfull boxes, and the final two auxiliary states must agree.

The fixed epoch is 4 October 2026 at 00:00 UTC. Archive members have sorted names, fixed timestamps and permissions, and stored compression. A closed manifest records every distributed non-manifest file with its byte count and SHA-256. The separately delivered TeX and PDF are byte-identical to the archive copies. With the same Python/TeX stack, clean extractions rebuild byte-identical artifacts; different TeX or font versions may change PDF bytes. Hashes detect accidental modification, not replacement of both the software and its hashes by an adversary. Full command instructions and the precise manifest schema are in `README_REPRODUCIBILITY.md`.

[write] 7 October 2026. In this report the package’s programs are `code/192-airy-*.py`, its fixtures, generated coefficients and recorded outputs `data/192-airy-*`, and its `README_REPRODUCIBILITY.md`, `DATA_SOURCES.md`, `code/README.md` and `data/README.md` are `192-airy-*.md`. The delivered PDF, the package `README.md` (replaced by this report’s) and `SHA256SUMS.json` are not shipped; the replay and build scripts check the closed delivered inventory, so they run only in a re-extracted archive. The report’s `README` gives a route that runs the exact checker on a copy.

The optional `diagnose_mpmath.py` evaluates finite towers and Airy expressions at chosen working precision. It supplies illustrations only, not interval certificates, certified decimal digits, effective remainders, or proof of any limiting assertion. In particular a short-range fit cannot establish the small amplitude difference in (7).

14 Prior sources and remaining questions

14.1 What is already part of the literature

Hanna’s A096537 entry defines the linear continued exponential, and A096542 explicitly gives the entire shifted equation (2), not merely its specialization at $y = 1$ [1, 2]. Neither the sequence nor that family is introduced here as new.

The primary metadata and abstract of Bender and Vinson’s 1996 paper describe a general continued-exponential representation with coefficients at each level and a tree-graph interpretation carrying the corresponding vertex amplitudes [3]. Thus the broad tower/tree framework deserves prior credit. The specialization of its general displayed tower to level coefficients $1, 2, 3, \dots$ is immediate algebra; it does not reveal the scope of results in the unavailable article body. A full-text theorem comparison with that paper has not been completed here.

Poland’s 1998 paper treats continued exponentials in statistical mechanics [4]. Publisher metadata identifies it as open-archive, but its full text could not be retrieved in this environment. This is an access limitation of this comparison, not evidence that the paper is closed or irrelevant. In particular, the exact overlap with linear unbounded weights, coefficient asymptotics, or Airy finite-height behavior remains unresolved for both papers.

Chapoton’s inspected paper concerns hyperplane arrangements associated with fixed labeled rooted trees. Its depth-product characteristic polynomial and chamber-count formulas are relevant related enumerative structure [5]. Its later lattice enumeration is attached to a fixed tree; no summed coefficient asymptotic of the kind proved here was found in the inspected text. This is a statement about that paper, not its surrounding literature. Similarly named sequential depth-

weighted attachment models should not be identified with the static probability proportional to (17) without a separate equivalence argument.

A bounded search using the sequence identifiers, the proposed corrected constant and decimal variants, and continued-exponential/Airy/depth-weighted terms found no exact prior occurrence of the correction or the specific finite-height theorem. Formula indexing and full-text access are incomplete. That negative search result is not a priority certificate. The defensible conclusion here is the proved correction to the currently displayed OEIS amplitude, with prior-method attribution and explicit limits on literature coverage. The primary special-function inputs are the standard Airy facts listed in [6, 7, 8]; the finite-tower, Fourier-tail, and coefficient-transfer arguments are given in full in this report.

14.2 Questions not settled by the present estimates

The following are promising extensions, not conclusions of this report.

1. **Rates and further coefficients.** The entrance and matching arguments establish uniform equivalents in the iterated-limit form required here. They do not give an effective rate for the final coefficient equivalent. A further term would require quantitative entrance and sensitivity errors, a refined real or complex window, a corresponding quantitative Fourier L^1 remainder, and propagation through the height sum. Formal expansions or fitted powers alone do not supply those estimates.
2. **Fixed positive shifts.** The shifted functional equation is already prior. Determining a complete y -dependent polynomial exponent and amplitude needs a separate normalization and uniform analysis; checks of finite polynomial rows do not prove it.
3. **Larger tilt domains and rare heights.** The compact real window used here stops at the convenient upper bound 1. Extending it toward Airy zeros, or obtaining full height large-deviation equivalents away from the central range, requires additional analysis.
4. **Effective inversion.** The shrinking sandwich is an asymptotic existence result. A usable certified threshold algorithm with a proved error bar would need explicit constants and an onset bound for the coefficient error, followed by a crossing-aware integer comparison.
5. **Full primary-source comparison.** Authorized full-text copies of Bender–Vinson and Poland would permit a page-level comparison of variable weights, divergent formal series, critical finite depth, and coefficient asymptotics. Until then no absolute novelty assertion is warranted.

The central distinction is between a formal Airy balance and a coefficient theorem. Here the normalized entrance, real-window continuation, separate minor arcs, uniform lattice inversion, and all-height domination provide the complete bridge. Further-order work must preserve all of those controls.

[write] **7 October 2026.** Part II (Report 193) answers the first-coefficient part of item 1, with $b_n \rho^n \sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3})$, $-0.61 < c_1 < -0.25$ (Theorem 15.1), and item 2 for every fixed positive shift, with the Gamma prefactor uniformly on compact sets of $y > 0$ (Theorem 15.2). Effective rates, further coefficients, shifts near 0 or growing, and items 3–5 remain open; all are gathered in Section 29.

Part II

Fractional corrections and Gamma laws for exponential towers: a sequel to the leading Airy asymptotic

Bundle Report 193, archive A096537_Fractional_Corrections_and_Gamma_Laws_Source.zip, delivered as Report193.tex (29 pages). Delivered title block “Report193 / Fractional Corrections and Gamma Laws for Exponential Towers / A sequel to the leading Airy asymptotic / 4 October 2026”. Printed as delivered, with [write] notes; Report 193’s Section k is Section $k + 14$ here and its statements move with their sections; its equation (k) is equation $(k + 109)$ here. The opening paragraph “Prior constructions and scope” is the manuscript’s own. Its Report192 is Part I.

Abstract of this Part (as delivered). For the known formal family $F_y(x) = \exp(xyF_{y+1}(x))$, put $P_n(y) = n![x^n]F_y(x)$ and $\rho = 1 - e^{-1}$. We prove that $P_n(1)\rho^n\sqrt{n}/(n!)^2 = C + c_1n^{-1/3} + o(n^{-1/3})$, where $C = \sqrt{2/(\pi\rho)}$ and $-0.61 < c_1 < -0.25$. The coefficient is defined by an absolutely convergent Airy perturbation integral. Its strict sign follows from a real-Airy reduction, a proved positive-real tail bound, and reproducible directed interval quadrature. The discrete proof retains the parabolic entrance logarithm, controls a growing complex window, and cancels the larger overlap terms before Fourier inversion and height averaging. We also prove the same relative fractional correction uniformly on compact positive real y sets, together with the Gamma prefactor $P_n(y)/(n!)^2 \sim C\rho^{1-y}\rho^{-n}n^{y-3/2}/\Gamma(y)$, a normal limit for root-ancestor choices, and refined threshold inverses. The first correction gives eventual approach from below, but no effective finite-index onset or monotonicity of the normalized sequence. The report makes no all-orders, complex Borel-sector, or absolute-priority claim.

Prior constructions and scope. The shifted family and polynomial triangle are due to Paul D. Hanna as recorded in OEIS A096537 and A096542 [1, 2]. General continued exponentials and their tree interpretation belong to the earlier framework of Bender and Vinson [3]. A complete full-text theorem comparison with that paper and Poland’s continuation [4] remains unresolved. The formulas proved here are not presented as a worldwide novelty claim. Report192 established the leading unshifted Airy coefficient theorem; Section 16 states exactly the leading input retained here. The new error estimates, sign reduction, and offset normalization are proved in the present text rather than inferred from fitted coefficients.

15 Statements and normalizations

All logarithms are natural. Set

$$\alpha = e - 1, \quad \rho = \alpha/e, \quad a = 2^{-1/3}, \quad C = \sqrt{\frac{2}{\pi\rho}}, \quad \kappa = 2^{1/6}\sqrt{\frac{e}{\pi}}. \quad (110)$$

The equations

$$F_y(x) = \exp(xyF_{y+1}(x)), \quad F_y(0) = 1 \quad (111)$$

define a unique formal family: degree n on the right needs degrees only through $n - 1$ at the next level. We write

$$P_n(y) = n![x^n]F_y(x), \quad b_n(y) = \frac{P_n(y)}{(n!)^2}, \quad a_n = P_n(1), \quad b_n = b_n(1).$$

No analytic convergence of an infinite positive-real tower is used. Every analytic tower evaluation below is finite and entire.

Theorem 15.1 (The first fractional correction). *As $n \rightarrow \infty$,*

$$b_n \rho^n \sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3}), \quad -0.61 < c_1 < -0.25. \quad (112)$$

The constant c_1 is the absolutely convergent integral in (121), equivalently the real integral in (185). Thus $b_n \rho^n \sqrt{n} < C$ for all sufficiently large n , with a gap of order $n^{-1/3}$.

The bracket is a rigorous enclosure under the stated arithmetic trust boundary, not a certified decimal expansion. The ordinary numerical value -0.43652529 is only a diagnostic. No finite onset of the strict inequality, no monotonicity of $b_n \rho^n \sqrt{n}$, and no explicit numerical error bound at a specified finite n is asserted.

Theorem 15.2 (The Gamma prefactor). *For every compact $Y \subset (0, \infty)$,*

$$b_n(y) = C_y \rho^{-n} n^{y-3/2} \left[1 + d n^{-1/3} + o_Y(n^{-1/3}) \right], \quad y \in Y, \quad (113)$$

$$C_y = \frac{C \rho^{1-y}}{\Gamma(y)}, \quad d = \frac{c_1}{C}.$$

The relative coefficient d is independent of y . The same critical Airy size law applies to every fixed positive shift. For the root-ancestor statistic K_n defined in Section 25, with $L_n = \log(n/\rho)$,

$$\frac{P_n(y)}{P_n(1)} = \frac{(n/\rho)^{y-1}}{\Gamma(y)} [1 + o_Y(n^{-1/3})], \quad \frac{K_n - L_n}{\sqrt{L_n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (114)$$

The ratio equivalent is uniform on every such Y .

Uniformity near $y = 0$ and shifts growing with n are outside the theorem. A separate quantitative local complex argument in Section 25.1 justifies mean and variance constants; these are not obtained by differentiating a merely real asymptotic.

[write] **7 October 2026.** At $y = 1$, $C_y = C$ and Theorem 15.2 is Part I's Theorem 1.1 with the correction of Theorem 15.1 added. The theorem answers Part I's question "Fixed positive shifts" (Section 14) for every fixed $y > 0$.

15.1 Definition of the correction without numerical fitting

For $\lambda \in i\mathbb{R}$ let

$$w''_\lambda(t) = \frac{1}{2}(t - \lambda)w_\lambda(t), \quad w_\lambda(0) = 0, \quad w'_\lambda(0) = 1, \quad (115)$$

$$U_\lambda(t) = 2w'_\lambda(t)/w_\lambda(t), \quad G_\lambda(t) = \frac{1}{2} - \frac{1}{2}(t - \lambda)U_\lambda(t) - \frac{1}{12}U_\lambda(t)^3. \quad (116)$$

The nonvanishing needed here is proved in Section 18. For any constant c , define

$$V_c(t) = \frac{1}{w_\lambda(t)^2} \left\{ c - \frac{2}{3} \log t + \int_0^t \left[w_\lambda(r)^2 G_\lambda(r) + \frac{2}{3r} \right] dr \right\}. \quad (117)$$

The integral is convergent, and $V'_c + U_\lambda V_c = G_\lambda$. Let $I'_c = V_c$ have the uniquely fixed finite-part normalization

$$I_c(t) - \frac{\frac{2}{3}(\log t + 1) - c}{t} \rightarrow 0 \quad (t \downarrow 0), \quad (118)$$

and set, with the principal square root,

$$J_c(\lambda) = \lim_{T \rightarrow \infty} \left[I_c(T) + \frac{(T - \lambda)^2}{3} - \frac{4}{3} \sqrt{2(T - \lambda)} \right]. \quad (119)$$

Existence and polynomial growth on the imaginary axis are established below. Put

$$\mathcal{F}(\lambda) = \frac{\kappa}{\text{Ai}(-a\lambda)^2}, \quad \mathcal{Q}(\lambda) = \mathcal{F}(\lambda) \left[\frac{\lambda^2}{2} - J_0(\lambda) \right]. \quad (120)$$

Then the exact first correction is

$$c_1 = \frac{\alpha^{-1/6}}{2\pi} \int_{\mathbb{R}} \mathcal{Q}(i\tau) d\tau. \quad (121)$$

Conjugation makes this integral real. The definition contains neither fitted finite-sequence data nor an arbitrary normalization constant.

16 Exact heights and the leading input

For $y > 0$, write $\beta = y - 1$, $\ell_j = j + \beta$, and define

$$A_{h+1,h;y} = 1, \quad A_{j,h;y}(x) = \exp(\ell_j x A_{j+1,h;y}(x)), \quad E_{h,y} = A_{1,h;y} - A_{1,h-1;y}. \quad (122)$$

The second tower is 1 at level h ; $A_{1,0;y} = 1$. Every $E_{h,y}$ has nonnegative coefficients and least degree h : at the bottom the difference is $e^{\ell_h x} - 1$, and each upward step multiplies the difference by x before applying $e^u - e^v = e^v(e^{u-v} - 1)$. Thus, with $c_{n,h;y} = [x^n]E_{h,y}$,

$$b_n(y) = \sum_{h=1}^n \frac{c_{n,h;y}}{n!} \quad (n \geq 1). \quad (123)$$

For $y = 1$ we abbreviate $E_{h,1} = H_h$ and $c_{n,h;1} = c_{n,h}$.

Leading-input recap. The unshifted leading result from Report192 is

$$H_h((eh)^{-1}) \sim \mathcal{F}(0)h^{1/6}e^{-\alpha h}, \quad (124)$$

$$c_{n,h}/n! \leq B \frac{e^n}{n!} h^{n-1/2} e^{-\alpha h} \quad (h \geq 1, n \geq 0), \quad (125)$$

for an absolute B . The centered finite-height characteristic transform has a bound $Be^{-b|t|^{3/2}}$ beyond a fixed central interval under $\theta = h^{-2/3}t$, until the remote arc, whose contribution is exponentially small in h . The Airy inverse

$$f(z) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itz} \mathcal{F}(it) dt$$

has bounded derivatives of every fixed order and $f(0) = \kappa/a = \sqrt{2e/\pi}$. This recap is a leading theorem, not a first-order remainder assumption. The present argument also recovers these leading facts from its quantitative window and the independent positive factorization below, so they can be read as orientation rather than separate assumptions. Sections 17–22 prove the additional quantitative estimates needed for Theorem 15.1. For completeness, the positive-factor minor-arc argument and its shifted version are given in Section 21; Sections 20–22 give the full normalization and height sum underlying Theorem 15.2.

[write] 7 October 2026. Report192 is Part I. The recap is Part I's Theorem 5.1 (with $\mathcal{F}(0) = \kappa/\text{Ai}(0)^2 = K$), its all-height bound (95), its minor-arc Theorem 8.1 and its density, here as $f = Kp$ with p of (74). Where this Part re-derives these facts inside its quantitative argument, the proofs are printed in full, as their estimates are needed for the correction.

Airy facts used. The Wronskian and connection identities [6] give

$$w_\lambda(t) = \frac{\pi}{a} [\text{Ai}(-a\lambda) \text{Bi}(a(t - \lambda)) - \text{Bi}(-a\lambda) \text{Ai}(a(t - \lambda))]. \quad (126)$$

We use the separate function and derivative sector expansions, with remainders, from [7]; asymptotic equivalents are never differentiated without those derivative expansions. The zeros of Ai are negative real [8]. On the imaginary axis the function $\mathcal{F}$, its fixed-order derivatives, and its products with fixed polynomials are bounded by a polynomial times $e^{-b|\tau|^{3/2}}$. Finally, if $r(\lambda) = \text{Bi}(-a\lambda)/\text{Ai}(-a\lambda)$, then

$$r'(\lambda) = -\frac{a}{\pi\kappa} \mathcal{F}(\lambda), \quad r(i\infty) = -i, \quad r(-i\infty) = i.$$

Integrating the derivative upward proves $f(0) = \kappa/a$ and also shows that r is bounded on the imaginary axis.

17 The normalized parabolic entrance

Set $\epsilon = h^{-1/3}$ and evaluate at $x = e^{\lambda\epsilon^2}/(eh)$, where $\lambda = i\tau$. Use exponent variables $q_j^A = jxA_{j+1,h}(x)$ and $q_j^B = jxA_{j+1,h-1}(x)$, so no complex logarithm is needed to define them. With $u_k^p = 1 - q_{h-k}^p$, the exact branches are

$$u_k^p = 1 - e^{\lambda\epsilon^2}(1 - k/h)e^{-u_{k-1}^p}, \quad u_0^A = 1 - e^{\lambda\epsilon^2}/e, \quad u_0^B = 1. \quad (127)$$

Write $\bar{u}_k = (u_k^A + u_k^B)/2$ and $\Delta_k = u_k^B - u_k^A$.

Lemma 17.1 (Autonomous normalization). *Let $v_0 = 1$ and $v_{k+1} = 1 - e^{-v_k}$. There is a finite constant c_* such that*

$$v_k = \frac{2}{k} - \frac{\frac{2}{3} \log k + 2c_*}{k^2} + \mathcal{O}(\log^2 k/k^3), \quad (128)$$

$$\frac{v_k + v_{k+1}}{2} = \frac{2}{k} - \frac{\frac{2}{3} \log k + 2c_* + 1}{k^2} + \mathcal{O}(\log^2 k/k^3), \quad (129)$$

$$v_k - v_{k+1} = \frac{2}{k^2} - \frac{\frac{4}{3} \log k + 4c_* + \frac{4}{3}}{k^3} + \mathcal{O}(\log^2 k/k^4). \quad (130)$$

Proof. The decreasing positive sequence tends to zero. For $r_k = 2/v_k$, Taylor's theorem gives $r_{k+1} = r_k + 1 + 1/(3r_k) + \mathcal{O}(r_k^{-3})$. First $r_k \sim k$, then summation gives $r_k = k + \mathcal{O}(\log k)$. The increments of $r_k - k - \frac{1}{3} \log k$ are therefore $\mathcal{O}(\log k/k^2)$, an absolutely summable sequence. Its tail is $\mathcal{O}(\log k/k)$, proving convergence to c_* and (128). For the difference use $v_k - v_{k+1} = v_k^2/2 - v_k^3/6 + \mathcal{O}(v_k^4)$ rather than subtracting error terms. The midpoint follows by subtracting half this difference. $\square$

Take the quantitative cutoffs

$$K = \lceil \epsilon^{-3/5} \rceil, \quad t_0 = \epsilon K, \quad M = \lceil \epsilon^{-11/8} \rceil, \quad T = \epsilon M, \quad i = h - M, \quad (131)$$

and the growing Fourier window

$$|\tau| \leq L_\epsilon = A(\log(1/\epsilon))^{2/3}. \quad (132)$$

Here A is any fixed positive constant, chosen later large enough for the Fourier tails. In the remainder estimates below, a factor called $\text{polylog}(1/\epsilon)$ means a fixed positive polynomial in $1 + \log(1/\epsilon)$, with degree independent of h, λ . Every parameter-dependent constant is polynomial in L_ϵ ; no uncontrolled exponential of L_ϵ is allowed.

Lemma 17.2 (Early perturbation and separation). *Uniformly on (132),*

$$\bar{u}_K = \frac{v_K + v_{K+1}}{2} - \lambda\epsilon^2 K/3 + \mathcal{O}(L_\epsilon \epsilon^2 \log K + \epsilon^3 K^2 + L_\epsilon^2 \epsilon^4 K^3), \quad (133)$$

$$\log \Delta_K = \log 2 - 2 \log K - \frac{\frac{2}{3} \log K + 2c_* + \frac{2}{3}}{K} + \frac{\lambda\epsilon^2 K^2}{6} + \mathcal{O}(\epsilon^{6/5} \text{polylog}(1/\epsilon)). \quad (134)$$

The logarithm is continued from the positive real value at $\lambda = 0$.

Proof. Put $v_k^B = v_k$, $v_k^A = v_{k+1}$, $d_k^p = u_k^p - v_k^p$, and $b_k^p = e^{-v_{k-1}^p}$. Lemma 17.1 gives the summable kernel

$$\prod_{r=j+1}^k b_r^p \leq B \left(\frac{j+1}{k+1} \right)^2. \quad (135)$$

Writing $a_k = e^{\lambda\epsilon^2}(1 - k\epsilon^3)$, Taylor's theorem yields

$$d_k = b_k(d_{k-1} - (a_k - 1)) + \mathcal{O}(|a_k - 1||d_{k-1}| + |d_{k-1}|^2).$$

The kernel proves by bootstrap $|d_k| \leq B(|\lambda|\epsilon^2(k+1) + \epsilon^3(k+1)^2)$ for $k \leq K$: the nonlinear contribution relative to this bound is $\mathcal{O}((1+|\lambda|)\epsilon^2 K^2 + \epsilon^3 K^3) = o(1)$. The initial errors are $d_0^B = 0$ and $d_0^A = -\lambda\epsilon^2/e + \mathcal{O}(|\lambda|^2\epsilon^4)$. The linear response $l_k = b_k(l_{k-1} - 1)$, with $l_0^B = 0, l_0^A = -1/e$, satisfies $l_k = -k/3 + \mathcal{O}(\log(k+2))$: substituting $-k/3$ gives residual $\mathcal{O}(\log(k+2)/(k+1))$, and (135) sums it. Subtract $\lambda\epsilon^2 l_k$ and sum the remaining forcing to obtain

$$d_k = -\lambda\epsilon^2 k/3 + \mathcal{O}(|\lambda|\epsilon^2 \log(k+2) + \epsilon^3(k+1)^2 + |\lambda|^2\epsilon^4(k+1)^3).$$

In particular $\sum_{k < K} \bar{d}_k = -\lambda\epsilon^2 K(K-1)/6 + \mathcal{O}(\epsilon^{6/5} \text{polylog})$.

For $r(z) = \log(\sinh(z/2)/(z/2))$ the exact identity is

$$\log \Delta_K = \log \Delta_0 + \sum_{j=1}^K \log a_j - \sum_{k=0}^{K-1} \bar{u}_k + \sum_{k=0}^{K-1} r(\Delta_k). \quad (136)$$

Let $\Delta_k^* = v_k - v_{k+1}$. The same bootstrap gives $|\log(\Delta_k/\Delta_k^*)| \leq B(|\lambda|\epsilon^2(k+1)^2 + \epsilon^3(k+1)^3)$. Since $r'(z) = \mathcal{O}(|z|)$ in a fixed neighborhood of this trajectory and $\Delta_k^* = \mathcal{O}((k+1)^{-2})$, the difference of the r sums is at most $B(|\lambda|\epsilon^2 + \epsilon^3 \log(K+2))$. Subtract the autonomous identity in (136). The initial logarithm and $\sum \log a_j$ cost only $\mathcal{O}(|\lambda|\epsilon^2 K + \epsilon^3 K^2)$. The midpoint sum provides the displayed tilt; expanding $\log \Delta_K^*$ with Lemma 17.1 proves (134). All errors stated here have at least the $\epsilon^{6/5}$ margin up to fixed logarithmic factors. $\square$

The forced entrance parameter is

$$c_h = -\frac{2}{3} \log(1/\epsilon) - 2c_* - 1. \quad (137)$$

It is essential at finite height. Its eventual disappearance from the coefficient correction is proved by Fourier inversion, not assumed.

18 Complex stability and quantitative inner propagation

Lemma 18.1 (Energy stability on the whole imaginary axis). *For $\lambda = i\tau$, $\tau \in \mathbb{R}$, the solution $w_\lambda(t)$ is nonzero for $t > 0$, and*

$$\Re U_\lambda(t) \geq 2/t, \quad \Re U_\lambda(t) \geq c\sqrt{t} \quad (t \geq 1), \quad \Re U_\lambda(t) \geq U_0(t), \quad (138)$$

$$|U_\lambda(t) - U_0(t)| \leq |\tau|t/3, \quad |U_\lambda(t) - U_0(t)| \leq B|\tau|/\sqrt{t} \quad (t \geq 1). \quad (139)$$

All constants are absolute.

Proof. Integrate $w''\bar{w}$ and take real parts:

$$\Re(w'(t)\overline{w(t)}) = \int_0^t \left(|w'(s)|^2 + \frac{s}{2}|w(s)|^2 \right) ds. \quad (140)$$

A positive zero would contradict $w'(0) = 1$. Cauchy–Schwarz gives $\int_0^t |w'|^2 \geq |w(t)|^2/t$, proving the first bound. For $t \geq 2$, restrict the integral to $[t/2, t]$ and use the trace inequality

$$\int_A^B (|v'|^2 + q^2|v|^2) ds \geq q \tanh(q(B-A))|v(B)|^2.$$

To prove it, subtract $v(B) \cosh(q(s-A))/\cosh(q(B-A))$ and integrate the cross term by parts; the remainder energy is nonnegative. Taking $q = \sqrt{t}/2$ proves the second bound; $1 \leq t \leq 2$ follows from the first. Write $U = R + iI$. Its Riccati equation gives

$$(R - U_0)' = -\frac{1}{2}(R + U_0)(R - U_0) + \frac{1}{2}I^2.$$

Variation of constants from $\delta > 0$ and then $\delta \downarrow 0$ proves $R \geq U_0$: the initial discrepancy is $\mathcal{O}_\tau(\delta^3)$ and its propagator is at most $(\delta/t)^2$. Lastly $D = U_\lambda - U_0$ satisfies

$$D(t) = -i\tau \int_0^t \exp \left[-\frac{1}{2} \int_s^t (U_\lambda(r) + U_0(r)) dr \right] ds.$$

The kernel modulus is at most $(s/t)^2$, giving $|\tau|t/3$. For $t \geq 2$, split at $t/2$ and apply the square-root damping to bound its integral by $B/\sqrt{t}$. The compact remaining interval is immediate. $\square$

Integration of $\Re U \geq 2/t$ also gives $|w(t)| \geq t$. For the discrete towers, positivity of their exponent coefficients implies

$$|q_j(x_0 e^{i\theta})| \leq q_j(x_0), \quad \Re(1 - q_j(x_0 e^{i\theta})) \geq 1 - q_j(x_0), \quad x_0 = (eh)^{-1}. \quad (141)$$

Thus a complex tilt does not produce an exponential Gronwall loss.

Lemma 18.2 (Summable inner error). *With (131)–(137), set $a_k = \epsilon U_\lambda(\epsilon k) + \epsilon^2 V_{c_h}(\epsilon k)$. Uniformly on (132),*

$$\sum_{k=K}^M |\bar{u}_k - a_k| = \mathcal{O}(\epsilon^{17/16} \text{polylog}(1/\epsilon)), \quad (142)$$

$$|\bar{u}_M - a_M| = \mathcal{O}(\epsilon^3 T^{3/2} \text{polylog}(1/\epsilon)), \quad (143)$$

$$\log \Delta_M = \log \frac{2\epsilon^2}{w_\lambda(T)^2} + \epsilon D_{c_h}(T) + \mathcal{O}(\epsilon^{17/16} \text{polylog}(1/\epsilon)), \quad (144)$$

$$D_c(t) = -I_c(t) - t^2/2 + \lambda t + U_\lambda(t)/2. \quad (145)$$

The value $T = \epsilon M$ includes the integer rounding exactly.

Proof. We give the residual and kernel estimates, since a fixed-cutoff $o(1)$ would not suffice. On $0 < t \leq 1$, put $B_0 = -(2/3)\log t/t^2$ and $d = U - 2/t$. From Lemma 18.1, $|d| \leq B(1 + |\tau|)t$. Expanding $U^3 - 8/t^3$ shows

$$|G - (B'_0 + UB_0)| \leq B(1 + |\tau|)^3(1 + |\log t|)/t.$$

The linear V kernel has modulus at most $(s/t)^2$, so $|V_0 - B_0| \leq B(1 + |\tau|)^3(1 + |\log t|)$. Add c_h/w^2 . On $t \geq 1$ the same kernel decays at least as $e^{-c(t^{3/2}-s^{3/2})}$ and $|G| \leq P(1 + |\tau|)t^{3/2}$. It follows that

$$|V_{c_h}(t)| \leq P(1 + |\tau|) \left[\frac{1 + |c_h| + |\log t|}{t^2} + 1 + t \right]. \quad (146)$$

Here and below P may change, but is a fixed polynomial. Finite differentiation of the exact differential equations gives

$$|U'''| \leq P(1 + |\tau|)t^{-4}, \quad |V''| \leq P(1 + |\tau|)(1 + |c_h| + |\log t|)t^{-4} \quad (t \leq 1).$$

For $t \geq 1$, $U_0 = \mathcal{O}(\sqrt{t})$ and $U'_0 = \mathcal{O}(1)$ follow from the separate Airy expansions and compactness. The difference equation gives $|U'| \leq P(1 + |\tau|)$, hence $|U'''| \leq P(1 + |\tau|)t$; the V equation gives $|V'| \leq P(1 + |\tau|)(1 + |c_h|)t^{3/2}$ and $|V''| \leq P(1 + |\tau|)(1 + |c_h|)t^2$. The factor $1 + |c_h|$ is absorbed in the fixed polylogarithmic factor in every residual estimate. More precisely, the homogeneous term and its first two derivatives on $t \geq 1$ are bounded by $|c_h|P(1 + |\tau|)P(t)e^{-ct^{3/2}}$; this is also bounded by $|c_h|P(1 + |\tau|)t^{-4}$ after enlarging constants, as needed in the displayed residual bound.

Taylor expansion of a_{k-1} through ϵ^3 cancels the coefficients of ϵ^2 and ϵ^3 , because

$$U' = t - \lambda - U^2/2, \quad V' + UV = U''/2 - U^3/3 = G.$$

Thus the scalar recurrence residual at $t = \epsilon k$ is bounded by

$$P(1 + |\tau|)\epsilon^4 \left[\frac{(1 + |c_h| + |\log t|)^2}{t^4} + 1 + t^2 \right]. \quad (147)$$

The exact midpoint recurrence adds a factor $\cosh(\Delta_{k-1}/2)$. Coefficient positivity and the more strongly contracted critical-real pair give $|\Delta_k(i\tau)| \leq \Delta_k(0) \leq v_k - v_{k+1} \leq B/(k+1)^2$. Its extra residual is $\mathcal{O}(\epsilon^4/t^4)$ and is already included. The same bound ensures that the sinh quotient defining the separation logarithm has no zero along the chosen continuation.

For $K \leq k \leq \epsilon^{-1}$, the exact critical-real midpoint is at least $v_{k+1} \geq 3/(2(k+1))$ for sufficiently large K . The approximation has the same lower real-part bound: (146) gives $\epsilon|V|/\Re U = \mathcal{O}(\epsilon^{3/5} \text{polylog})$ on $[t_0, 1]$. Future propagation kernels therefore have total mass $\mathcal{O}(k)$. For later indices use the explicit barrier $b_k = \frac{1}{4}\sqrt{k/h}$. For $0 \leq u \leq 1$ and $0 \leq s \leq 1$,

$$1 - (1 - s)e^{-u} \geq u + s - su - u^2/2.$$

For $k \geq \lceil h^{1/3} \rceil$ and $k \leq M$, $b_k \leq 1/10$ eventually, so the right side at $u = b_{k-1}, s = k/h$ exceeds $b_{k-1} + (4/5)k/h$. Since $b_k - b_{k-1} \leq 1/(4\sqrt{hk}) \leq k/(4h)$, this is a subsolution. The autonomous starting bound proves it for both exact real branches. Equation (141) transfers it to their complex real parts. For the approximation, $\epsilon|V|/\Re U = \mathcal{O}(\epsilon\sqrt{T} \text{polylog}) = o(1)$, so its real part has the same contraction scale. Future kernel mass at $t_k \geq 1$ is $\mathcal{O}((\epsilon\sqrt{t_k})^{-1})$.

The early lemma and the small- t expansions

$$U = 2/t - \lambda t/3 + t^2/4 + \mathcal{O}(P(1 + |\lambda|)t^3), \quad V_c = \frac{c - (2/3)\log t}{t^2} + \mathcal{O}(P(1 + |\lambda|)(1 + |c| + |\log t|)) \quad (148)$$

give

$$K|\bar{u}_K - a_K| \leq P(1 + |\lambda|)[\log^2 K/K^2 + \epsilon^2 K \log(1/\epsilon) + \epsilon^3 K^3 + \epsilon^4 K^4] = \mathcal{O}(\epsilon^{6/5} \text{polylog}).$$

Sum (147) against the kernel masses. The total error is at most this entrance error plus fixed polynomial factors times

$$\epsilon^2 \int_{t_0}^1 (1 + |c_h| + |\log t|)^2 t^{-3} dt + \epsilon^2 \int_1^T (1 + t^{3/2}) dt.$$

These are respectively $\mathcal{O}(\epsilon^{6/5} \text{polylog})$ and $\mathcal{O}(\epsilon^2 T^{5/2} \text{polylog}) = \mathcal{O}(\epsilon^{17/16} \text{polylog})$. For the endpoint, forcing before $T/2$ is superpolynomially damped, and the later residual divided by its local contraction gap is $\mathcal{O}(\epsilon^3 T^{3/2} \text{polylog})$.

Finally the exact multiplier is

$$\log(\Delta_k/\Delta_{k-1}) = \lambda\epsilon^2 + \log(1 - k/h) - \bar{u}_{k-1} + r(\Delta_{k-1}).$$

The r tail is $\mathcal{O}(K^{-3})$; the quadratic log remainder sums to $\mathcal{O}(\epsilon^3 T^3)$, both $o(\epsilon)$. Replace the midpoint by a_k using (142). Euler–Maclaurin yields its endpoint correction $+\epsilon U(T)/2$; the remainders are $\mathcal{O}(\epsilon^2 t_0^{-2} \text{polylog} + \epsilon^2 T)$. The leading integral is $2 \log w$. At t_0 , the early separation matches (145): its singular term is $[-(2/3) \log K - 2c_* - 2/3]/K$, and w supplies $\lambda\epsilon^2 K^2/6$. The omitted terms t_0^3 , $\lambda^2 t_0^4$, and $\epsilon t_0 \text{polylog}$ are $o(\epsilon)$ with the stated margin. This fixes the additive constant and proves (144). As a check, $D'_c = -V_c - (t - \lambda)/2 - U^2/4$ exactly. $\square$

19 Second outer correction and overlap cancellation

Let $\mathcal{T}(z)e^{-\mathcal{T}(z)} = z$ be the small tree branch and $v = 1 - \mathcal{T}$. Write $\eta = e^{\lambda\epsilon^2}/(eh)$, $z_j = j\eta$, and define

$$R = \frac{\mathcal{T}e^{\mathcal{T}}}{v^2}, \quad S = \frac{\mathcal{T}e^{2\mathcal{T}}(5 + v - v^2)}{2v^5}, \quad Q_{\text{out}} = \mathcal{T} + \eta R + \eta^2 S. \quad (149)$$

Direct differentiation verifies

$$S = \frac{\mathcal{T}}{v} \left[\frac{\mathcal{T}''}{2} + R' + \frac{(\mathcal{T}' + R)^2}{2} \right]. \quad (150)$$

All derivatives here are in z . This is the second cancellation in $Q_{\text{out}}(z) = ze^{Q_{\text{out}}(z+\eta)}$.

Lemma 19.1 (Summed outer control). *For either exact branch and $j \leq i = h - M$,*

$$|Q_{\text{out},j} - z_j e^{Q_{\text{out},j+1}}| \leq B|\eta|^3 |\mathcal{T}_j|/|v_j|^7, \quad (151)$$

$$\sum_{j \leq i} |q_j - Q_{\text{out},j}| \leq B \frac{|q_i - Q_{\text{out},i}|}{|v_i|} + B|\eta|^2 |v_i|^{-6} = o(\epsilon). \quad (152)$$

The error is uniform on (132).

Proof. Positivity of $\mathcal{T}$ and its derivative bound show

$$|\mathcal{T}(z_j) - \mathcal{T}(|z_j|)| \leq |\arg \eta| |z_j| \mathcal{T}'(|z_j|).$$

The relative perturbation of the real gap is $\mathcal{O}(L_\epsilon/T) = o(1)$. Thus all gaps lie in a fixed narrow cone and are comparable to the real ones, while $|\eta|/|v_i|^3 = \mathcal{O}(T^{-3/2})$. Analytic Taylor's theorem,

(150), and the derivative bounds for rational functions of $\mathcal{T}, v$ give (151). The actual q_i has modulus at most its real value, at most $1 - \epsilon\sqrt{T}$. The approximation lies in the same type of disk, since $\eta R/v = \mathcal{O}(T^{-3/2})$ and $\eta^2 S/v = \mathcal{O}(T^{-3})$. Choose a radius r between $\mathcal{T}(|z_i|)$ and 1 large enough for both. Then $|z_i|e^r \leq r$, so the disk is invariant under descent; each mean-value derivative has modulus at most $1 - \epsilon'|v_i|$. Summing the residuals along the ray, then the propagation kernel, gives (152); the residual contribution is $\mathcal{O}(T^{-3})$. The change of variable $dz = e^{-\mathcal{T}} v d\mathcal{T}$ justifies the powers in this integral estimate, and neighboring gaps are comparable because $|\eta|/|v_i|^2 = o(1)$.

For the incoming error put $s = T - \lambda$. The bounded connection ratio $r(\lambda)$ and the separate sector expansions give, uniformly,

$$U(T) = \sqrt{2s} - \frac{1}{2s} - \frac{5}{8\sqrt{2}s^{5/2}} + \mathcal{O}(s^{-4}), \quad V_{ch}(T) = -2s/3 + \mathcal{O}(\text{polylog}/\sqrt{s}). \quad (153)$$

Expansion of $v_i - \eta R_i - \eta^2 S_i$ gives exactly the three displayed ϵU terms and the leading $\epsilon^2 V$ term. Using (143), the incoming discrepancy is bounded by

$$\text{polylog}(1/\epsilon)[\epsilon T^{-4} + \epsilon^2 T^{-1/2} + \epsilon^3 T^{3/2}].$$

Division by $|v_i| \asymp \epsilon\sqrt{T}$ gives $\mathcal{O}(\text{polylog}[T^{-9/2} + \epsilon/T + \epsilon^2 T]) = o(\epsilon)$. The residual $T^{-3} = \mathcal{O}(\epsilon^{9/8})$ is also $o(\epsilon)$. $\square$

19.1 The equilibrium product at a moving endpoint

Let $P = \prod_{j=1}^{i-1} \mathcal{T}(z_j)$. The exact identity $\log(q_j/\mathcal{T}_j) = q_{j+1} - \mathcal{T}_j$ and Lemma 19.1 imply

$$\log \prod_{j < i} \frac{q_j}{\mathcal{T}_j} = -\log v_i + \eta \int_0^{z_i} S(z) dz + o(\epsilon). \quad (154)$$

Indeed $\int R dz = -\mathcal{T} - \log v$ cancels the endpoint $+\mathcal{T}_i$; the lower endpoint is $\mathcal{O}(\eta)$, the R discretization error is $\mathcal{O}(\epsilon/T)$, and the S error is $\mathcal{O}(\epsilon T^{-5/2})$. The separation at i is exponentially small in $T^{3/2}$, so its divided-difference product has the same logarithm as either individual branch product to the required accuracy.

An overlap term larger than the desired correction must be retained:

$$\eta \int_0^{z_i} S(z) dz = \frac{5}{6(2s)^{3/2}} + \mathcal{O}(\epsilon/T + T^{-3}), \quad (155)$$

$$\log w_\lambda(T)^{-2} = \log \frac{a^{5/2}}{\pi \text{Ai}(-a\lambda)^2} + \frac{1}{2} \log s - \frac{2\sqrt{2}}{3} s^{3/2} - \frac{5}{6(2s)^{3/2}} + \mathcal{O}(T^{-3}). \quad (156)$$

Each $s^{-3/2}$ term has size $\epsilon^{9/16}$, greater than ϵ . They cancel; discarding either would invalidate the proof.

Put $\hat{H} = he^{-\lambda\epsilon^2}$ and $r = i - 1$. Isolating $\log \mathcal{T}(z) = \log z + \mathcal{T}(z)$ permits exact factorial summation at zero. Stirling and trapezoidal Euler–Maclaurin yield

$$\log P = \hat{H} \int_0^{r/\hat{H}} \log \mathcal{T}(v/e) dv + \frac{1}{2} \log(2\pi r) + \frac{1}{2} \mathcal{T}(r/(e\hat{H})) + o(\epsilon). \quad (157)$$

For the smooth part the error is $\mathcal{O}(|\eta| \int |\mathcal{T}''| |dz|) = \mathcal{O}(|\eta|/|v_i|) = \mathcal{O}(\epsilon^2/\sqrt{T})$, by gap-cone comparability. This is a moving-endpoint bound, stronger than a fixed-endpoint $o(1)$. Since

$$\int_0^1 \log \mathcal{T}(v/e) dv = -\alpha, \quad \log \mathcal{T}((1-y)/e) = -\sqrt{2y} - y/3 + \mathcal{O}(y^{3/2}),$$

and

$$1 - r/\widehat{H} = \epsilon^2 s + \epsilon^3 + \epsilon^4(\lambda T - \lambda^2/2) + \mathcal{O}(\epsilon^5 \text{polylog}),$$

expansion of the removed endpoint integral gives

$$\begin{aligned} \log P = & -\alpha h + \alpha\lambda/\epsilon + \frac{1}{2}\log(2\pi e h) + \frac{2\sqrt{2}}{3}s^{3/2} \\ & + \epsilon \left[-\frac{\alpha\lambda^2}{2} + \frac{\sqrt{2}s}{2} + \frac{s^2}{6} \right] + \mathcal{O}(\epsilon^2 T^{5/2} \text{polylog}), \end{aligned} \quad (158)$$

$$-\log v_i = -\log \epsilon - \frac{1}{2}\log(2s) + \epsilon\sqrt{2s}/3 + \mathcal{O}(\epsilon^2 T \text{polylog}). \quad (159)$$

The ϵ^3 endpoint shift produces an essential order- ϵ term. The term $-\alpha\lambda^2/2$ comes from the second term of $e^{-\lambda\epsilon^2}$ and is equally essential.

19.2 Completion of the finite-height correction

The difference at level i , its outer product, and the final outer exponential give H_h . That final exponential changes the logarithm by $\mathcal{O}(h^{-1})$. Combining (144), (154)–(159) cancels the two larger overlap terms and gives leading factor $h^{1/6}\mathcal{F}(\lambda)$. The remaining order- ϵ coefficient is

$$-\frac{\alpha\lambda^2}{2} + \frac{\lambda^2}{2} - I_c(T) - \frac{s^2}{3} + \frac{5}{6}\sqrt{2s} + \frac{1}{2}U(T).$$

The V equation gives

$$I_c(T) = J_c(\lambda) - s^2/3 + (4/3)\sqrt{2s} + \mathcal{O}(\text{polylog } T). \quad (160)$$

Together with (153), all growing terms cancel. The Wronskian identity

$$\frac{d}{dt} \frac{\text{Ai}(a(t-\lambda))}{w_\lambda(t)} = -\frac{\text{Ai}(-a\lambda)}{w_\lambda(t)^2}$$

fixes the homogeneous finite part and proves

$$J_c - J_0 = ca \frac{\text{Ai}'(-a\lambda)}{\text{Ai}(-a\lambda)}, \quad -\mathcal{F}(J_c - J_0) = -\frac{c}{2}\mathcal{F}'(\lambda). \quad (161)$$

Consequently

$$\begin{aligned} h^{-1/6} e^{\alpha h - \alpha\lambda/\epsilon} H_h(e^{\lambda\epsilon^2}/(eh)) = & \mathcal{F}(\lambda) + \epsilon \left[\mathcal{Q}(\lambda) - \frac{\alpha}{2}\lambda^2\mathcal{F}(\lambda) - \frac{c_h}{2}\mathcal{F}'(\lambda) \right] \\ & + \mathcal{O}(\epsilon^{17/16} \text{polylog}(1/\epsilon)). \end{aligned} \quad (162)$$

The logarithmic correction after all cancellations is $\epsilon \text{polylog}$; exponentiation costs only $\mathcal{O}(\epsilon^2 \text{polylog})$. This proves an actual uniform expansion on the growing window, not a formal matching rule.

20 The Gamma factor on an offset grid

We now extend the quantitative proof, not merely its leading equivalent, to y in a fixed compact $Y \subset (0, \infty)$. Set

$$\beta = y - 1, \quad H = h + \beta, \quad \epsilon = H^{-1/3}, \quad x = e^{\lambda\epsilon^2}/(eH), \quad B_y = e^\beta/\Gamma(y).$$

At terminal distance k , $\ell_{h-k}/H = 1 - k/H$ exactly. Therefore the shifted recurrence is exactly (127), with H in place of h , up to its final integer index $h - 1 = H - y$. Its initial pair is unchanged. The inner cutoffs, ceilings of the real powers of ϵ , lie far before $H - y$, uniformly on Y . Every proof in Sections 17–18 uses integer k but never integrality of H . Positivity is preserved by $\ell_j > 0$. Thus all inner bounds and the same c_H hold uniformly on Y .

The outer grid $z_j = (j + \beta)\eta$, $\eta = e^{\lambda\epsilon^2}/(eH)$, has spacing exactly η . At $i = h - M$, its endpoint weight is $\ell_i = H - M$, so the incoming gap and overlap calculations are identical. The residual retains its factor $\mathcal{T}_j$ near the origin. Descending contraction uses only $|z_j| \leq |z_i|$, hence is uniform also when $0 < y < 1$. The exact product identity is

$$\sum_{j < i} \log(q_j/\mathcal{T}_j) = \sum_{j=2}^i (q_j - \mathcal{T}_j) + \mathcal{T}_i - \mathcal{T}_1.$$

Near zero, $\mathcal{T}, R, S = \mathcal{O}(z)$. The missing lower intervals in the integrals have length $\mathcal{O}_Y(|\eta|)$ and contribute $\mathcal{O}_Y(|\eta|^2)$; the term $-\mathcal{T}_1$ and final outer exponential contribute $\mathcal{O}_Y(|\eta|) = \mathcal{O}_Y(\epsilon^3)$, below first order.

The product of the linear factors gives the new constant exactly. Put $r = i - 1$, $R_* = r + \beta = H - M - 1$, and $S_* = He^{-\lambda\epsilon^2}$. Then

$$\log \prod_{j=1}^r \mathcal{T}(\ell_j/(eS_*)) = \log \Gamma(R_* + 1) - \log \Gamma(y) - r \log(eS_*) + \sum_{j=1}^r \mathcal{T}(\ell_j/(eS_*)). \quad (163)$$

Apply the trapezoidal formula on the actual positive grid beginning at $y\eta$, not a fictitious grid continuing below zero. Its second-derivative integral is $\mathcal{O}(1/|v_i|)$, so its remainder is $\mathcal{O}(|\eta|/|v_i|)$. The lower-endpoint contribution, for example, is

$$\frac{1}{2}\mathcal{T}(y\eta) - \eta^{-1} \int_0^{y\eta} \mathcal{T}(z)dz = \frac{1}{2}\eta y(1 - y) + \mathcal{O}_Y(|\eta|^2).$$

Stirling at the real R_* and (163) therefore give

$$\begin{aligned} \log \prod_{j=1}^r \mathcal{T}(\ell_j/(eS_*)) &= S_* \int_0^{R_*/S_*} \log \mathcal{T}(v/e)dv + \beta \log(eS_*) - \log \Gamma(y) \\ &\quad + \frac{1}{2} \log(2\pi R_*) + \frac{1}{2} \mathcal{T}(R_*/(eS_*)) + \mathcal{O}_Y(|\eta|/|v_i| + |\eta|). \end{aligned} \quad (164)$$

The endpoint $R_* = H - M - 1$ is exactly the endpoint used in (158). The only new term is

$$\beta \log(eS_*) - \log \Gamma(y) = \beta(1 + \log H) - \log \Gamma(y) - \beta\lambda\epsilon^2. \quad (165)$$

The first two terms give $B_y H^\beta$, and the last is $\mathcal{O}_Y(\epsilon^2 \text{polylog})$. In particular it is not an additional order- ϵ tilt term. The error $|\eta|/|v_i| = \mathcal{O}(\epsilon^2/\sqrt{T})$ and all lower-endpoint errors also fit below the $\epsilon^{17/16}$ budget.

We have proved the uniform shifted transform expansion

$$\begin{aligned} B_y^{-1} H^{-\beta-1/6} e^{\alpha H - \alpha\lambda/\epsilon} E_{h,y}(x) &= \mathcal{F}(\lambda) + \epsilon \left[\mathcal{Q}(\lambda) - \frac{\alpha}{2} \lambda^2 \mathcal{F}(\lambda) - \frac{c_H}{2} \mathcal{F}'(\lambda) \right] \\ &\quad + \mathcal{O}_Y(\epsilon^{17/16} \text{polylog}(1/\epsilon)). \end{aligned} \quad (166)$$

This proves the critical leading amplitude as well as its quantitative correction uniformly on Y . No complex- y statement is used.

21 Fourier tails and the full first order remainder

21.1 Positive factorization on every minor arc

Write the full and truncated shifted towers at a fixed height as A_j, B_j and $\Delta_j = A_j - B_j$, and put

$$D_j(w) = \int_0^1 \exp(\ell_j w s \Delta_{j+1}(w)) ds.$$

Repeated divided differences give the exact entire identity

$$E_{h,y}(w) = (e^{Hw} - 1)w^{h-1} \prod_{j=1}^{h-1} \ell_j B_j(w) D_j(w). \quad (167)$$

All nonmonomial factors have nonnegative Taylor coefficients. At $x_0 = (eH)^{-1}$ this implies

$$\left| \frac{E_{h,y}(x_0 e^{i\theta})}{E_{h,y}(x_0)} \right| \leq \prod_{j=1}^{h-1} \frac{|B_j(x_0 e^{i\theta})|}{B_j(x_0)}. \quad (168)$$

Let $T_0 = 0$, $T_{L+1}(z) = ze^{T_L(z)}$. The exponent $\ell_j w B_{j+1}(w)$ dominates $T_{h-j}(\ell_j w)$ coefficientwise, since every deeper weight is at least ℓ_j . Hence its contribution to negative logarithmic modulus is at least

$$D_L(z, \theta) = \sum_{k \geq 1} [z^k] T_L(z) z^k (1 - \cos(k\theta)), \quad L = h - j, \quad z = \ell_j / (eH).$$

For $\tau = \mathcal{T}(z)$ and $\delta = 1 - \tau$, the deficit $d_L = \tau - T_L(z)$ satisfies

$$d_L \leq (6/L)e^{-\delta L/2}, \quad D_L(z, \theta) \geq D_\infty(z, \theta) - 2d_L. \quad (169)$$

Indeed $d_{j+1} = \tau(1 - e^{-d_j}) \leq d_j - d_j^2/3$ first gives $d_j \leq 3/j$; use this through half the steps, then $d_{j+1} \leq \tau d_j$ through the other half.

For $0 < |\theta| \leq 1/8$, choose integer terminal distances $k = h - j \in [H|\theta|, 2H|\theta|]$. On them $\ell_j/H = 1 - k/H$ exactly, also for real noninteger H . The degree block $m \in [1/|\theta|, 2/|\theta|]$ has $1 - \cos(m\theta) \geq 1 - \cos 1$. Using $[z^m]\mathcal{T}(z) = m^{m-1}/m!$ and Stirling, the terms on this block are bounded below by a constant times $m^{-3/2}$: the extra factor $(1 - k/H)^m$ is bounded below because $k/H \leq 2|\theta|$. Thus $D_\infty \geq c_0 \sqrt{|\theta|}$. The gap identity $k/H = \int_0^\delta v e^v dv$ gives $\delta \asymp \sqrt{|\theta|}$ and $\delta L \geq cH|\theta|^{3/2}$. By (169), the finite deficit is at most half this damping once $H|\theta|^{3/2}$ exceeds a fixed constant. There are at least $H|\theta|/2$ selected indices. Consequently

$$\left| \frac{E_{h,y}(x_0 e^{i\theta})}{E_{h,y}(x_0)} \right| \leq e^{-cH|\theta|^{3/2}} \quad A_0 H^{-2/3} \leq |\theta| \leq 1/8. \quad (170)$$

For $1/8 \leq |\theta| \leq \pi$, use levels with $\ell_j/H \in [1/4, 1/2]$. Their degree-one terms alone give e^{-cH} . All constants are uniform on Y , because each chosen range contains the claimed number of valid levels for large h . These far arcs exclude any other lattice resonance.

21.2 Polynomial growth of the perturbation finite part

We justify (119) and the global integrability required in inversion. At $t \leq 1$, the singular subtraction used in (146) gives $|I_0(1)| + |V_0(1)| \leq B(1 + |\tau|)^3$. Set $S_0 = \max(S_1, 2(1 + |\tau|))$ for a fixed sufficiently large S_1 . On $[1, S_0]$, stable propagation gives $|V_0(t)| \leq B(1 + |\tau|)^3(1 + t)$, hence $|I_0(S_0)| \leq B(1 + |\tau|)^5$.

On $t \geq S_0$, the sector of $s = t - i\tau$ stays strictly inside $|\arg s| < \pi/3$. The bounded connection ratio in (126) ensures the Ai part is exponentially smaller than the Bi part, with absolute constants. Thus (153) holds uniformly. Substitution of

$$V_{\text{as}}(t) = -2s/3 + (2\sqrt{2}/3)s^{-1/2}$$

in the V equation leaves residual $\mathcal{O}(s^{-3/2})$ (leading magnitude $17/(12\sqrt{2})$). Variation of constants, split at $t/2$, gives

$$|V_0(t) - V_{\text{as}}(t)| \leq Bt^{-2} + B(1 + |\tau|)^4 e^{-c(t^{3/2} - S_0^{3/2})}.$$

Its integral converges. It proves (160) and, conservatively,

$$|J_0(i\tau)| \leq B(1 + |\tau|)^5. \quad (171)$$

Accordingly $\mathcal{Q}(i\tau)$ is absolutely integrable and has Airy exponential decay up to a polynomial. The same holds for $\mathcal{F}'$ and $\lambda^2 \mathcal{F}$.

21.3 Full line error and the local limit

Extend the normalized centered transform

$$\Psi_{h,y}(t) = B_y^{-1} H^{-\beta-1/6} e^{\alpha H} e^{-i\alpha H^{1/3} t} E_{h,y}(e^{itH^{-2/3}}/(eH))$$

by zero outside $|t| \leq \pi H^{2/3}$. Its normalization at zero is bounded above and below uniformly on Y , by (166). Thus (170) gives the bound $Be^{-c|t|^{3/2}}$ outside a fixed compact interval, plus the exponentially small remote arc. Choose the fixed A in (132) large enough that $cA^{3/2} > 1$ with a strict margin for all relevant decay constants. The tails of the finite transform and $\mathcal{F}$ then have integral $o_Y(\epsilon)$. The tails of $\epsilon \mathcal{Q}$ and $\epsilon c_H \mathcal{F}'$ are also $o_Y(\epsilon)$, including $c_H = \mathcal{O}(\log H)$. On the growing central window, the integrated error in (166) is still $\mathcal{O}_Y(\epsilon^{17/16} \text{polylog}) = o_Y(\epsilon)$. We have therefore proved

$$\int_{\mathbb{R}} \left| \Psi_{h,y}(t) - \mathcal{F}(it) - \epsilon \left[\mathcal{Q}(it) - \frac{\alpha}{2}(it)^2 \mathcal{F}(it) - \frac{c_H}{2} \mathcal{F}'(it) \right] \right| dt = o_Y(\epsilon). \quad (172)$$

This is the first-order L^1 estimate; a fixed Fourier cutoff could not prove it.

In particular, define the integer Boltzmann size $N_{h,y}$ by

$$\mathbb{P}(N_{h,y} = n) = c_{n,h,y} x_0^n / E_{h,y}(x_0).$$

The leading term of (172) implies the uniform local limit

$$\sup_{y \in Y} \sup_{n \in \mathbb{Z}} \left| H^{2/3} \mathbb{P}(N_{h,y} = n) - p \left(\frac{n - \alpha H}{H^{2/3}} \right) \right| \longrightarrow 0, \quad p = f/\mathcal{F}(0). \quad (173)$$

The limiting characteristic function is $\text{Ai}(0)^2 / \text{Ai}(-ait)^2$, and $p(0) = \text{Ai}(0)^2/a$. The bound follows directly from the L^1 error in Fourier inversion; characteristic convergence and continuity at zero identify the probability law. For every height, after enlarging a compact- Y constant to cover the finite initial heights,

$$\mathbb{P}(N_{h,y} = n) \leq B_Y H^{-2/3}, \quad \frac{c_{n,h,y}}{n!} \leq B_Y \frac{e^n}{n!} H^{n+\beta-1/2} e^{-\alpha H}. \quad (174)$$

For the finitely many initial heights this follows by continuity in y : H is bounded above and away from zero on Y , and the finite entire tower evaluation divided by its stated positive amplitude has a finite maximum. Thus this is a genuinely all-height majorant.

22 Height averaging and the universal relative correction

Let q be the inverse Fourier transform of $\mathcal{Q}$, a continuous function, and put $z = (n - \alpha H)/H^{2/3}$. Inverting (172) gives uniformly in integer n and $y \in Y$

$$\frac{c_{n,h;y}}{n!} = B_y \frac{e^n}{n!} H^{n+\beta-1/2} e^{-\alpha H} \left\{ f(z) + H^{-1/3} \left[q(z) - \frac{\alpha}{2} f''(z) - \frac{c_H}{2} z f(z) \right] + o_Y(H^{-1/3}) \right\}. \quad (175)$$

Integration by parts sends $\mathcal{F}'$ to $zf(z)$, while $\lambda^2 \mathcal{F}$ transforms to $f''(z)$; the Airy tails remove all boundary terms.

By (174), heights outside $|H - n/\alpha| \leq \sqrt{n} \log n$ are negligible relative to the full sum by more than any fixed inverse power. To check this uniformly, the logarithm of the positive weight $H^{n+\beta-1/2} e^{-\alpha H}$ has its maximum at $(n + \beta - 1/2)/\alpha$ and second derivative of order $-1/n$ near that point. Outside a fixed proportional central interval the exponent loses cn , and in its central complement it loses $c(H - n/\alpha)^2/n$. Integral comparison on each monotone tail transfers these bounds to the shifted grid $H = y, y + 1, \dots$. The retained heights satisfy $h \leq n$ for large n because $\alpha > 1$.

The full shifted-grid weight sum is

$$\sum_{h \geq 1} (h + \beta)^{n+\beta-1/2} e^{-\alpha(h+\beta)} = \frac{\Gamma(n + \beta + 1/2)}{\alpha^{n+\beta+1/2}} [1 + \mathcal{O}_Y(n^{-1/2})]. \quad (176)$$

Indeed a unimodal positive function differs in integral and grid sum by at most twice its maximum; the omitted interval $[0, y]$ is harmless. Uniform Stirling makes the maximum-to-integral ratio $\mathcal{O}_Y(n^{-1/2})$. This grid error is below the $n^{-1/3}$ scale.

Let $\mathbb{E}_w$ denote expectation under the normalized weight on the retained window. Direct Gamma integration and the same sum-integral comparison give

$$\mathbb{E}_w z = \mathcal{O}_Y(n^{-2/3}), \quad \mathbb{E}_w z^2 = \alpha^{4/3} n^{-1/3} + o_Y(n^{-1/3}), \quad (177)$$

$$\mathbb{E}_w |z|^3 = \mathcal{O}_Y(n^{-1/2}), \quad \mathbb{E}_w H^{-1/3} = \alpha^{1/3} n^{-1/3} (1 + o_Y(1)). \quad (178)$$

For clarity, the continuous first two moments are exactly

$$\alpha^{2/3} (1/6 - \beta) \frac{\Gamma(n + \beta - 1/6)}{\Gamma(n + \beta + 1/2)}, \quad (179)$$

$$\alpha^{4/3} \left[n + \beta^2 - \frac{2}{3}\beta - \frac{5}{36} \right] \frac{\Gamma(n + \beta - 5/6)}{\Gamma(n + \beta + 1/2)}. \quad (180)$$

For the signed first moment, split z times the weight at its zero and its finitely many extrema. Its maximum absolute value is $\mathcal{O}_Y(n^{-1/6})$ times the weight maximum, so its normalized lattice error is $\mathcal{O}_Y(n^{-2/3})$. The second-moment error is $\mathcal{O}_Y(n^{-5/6})$. Gaussian rescaling gives the absolute third moment; deleting the already bounded tails changes none of these orders. This explicitly rules out an order- $n^{-1/3}$ lattice-phase displacement.

Taylor expansion of f through second order, using its bounded third derivative, yields

$$\mathbb{E}_w f(z) = f(0) + \frac{1}{2} \alpha^{4/3} f''(0) n^{-1/3} + o_Y(n^{-1/3}).$$

On the shrinking z window, continuity of q, f'' gives

$$\mathbb{E}_w H^{-1/3} [q(z) - \frac{\alpha}{2} f''(z)] = \alpha^{1/3} [q(0) - \frac{\alpha}{2} f''(0)] n^{-1/3} + o_Y(n^{-1/3}).$$

The two f'' terms cancel exactly. The entrance contribution is bounded in absolute value by $\mathcal{O}_Y(n^{-1/2} \log n) = o_Y(n^{-1/3})$. Thus the averaged braces in (175) are

$$f(0) + \alpha^{1/3} q(0) n^{-1/3} + o_Y(n^{-1/3}). \quad (181)$$

Finally, (176), (123), and uniform Stirling give

$$B_y \frac{e^n \Gamma(n + \beta + 1/2)}{n! \alpha^{n+\beta+1/2}} = B_y \alpha^{-\beta-1/2} \rho^{-n} n^{\beta-1/2} [1 + \mathcal{O}_Y(1/n)].$$

Since

$$B_y \alpha^{-\beta-1/2} f(0) = C_y, \quad \frac{\alpha^{1/3} q(0)}{f(0)} = \frac{c_1}{C},$$

this proves the asymptotic parts of Theorems 15.1 and 15.2. The sign of c_1 is established next.

23 A real Airy integral for the correction

The complex perturbation integral is useful for deriving the coefficient; a different representation is better for certifying its sign. For $t > 0$ define

$$\begin{aligned} A &= \text{Ai}(at), & B &= \text{Bi}(at), & p &= (\pi/a)A(B + iA), \\ v &= 2a \text{Ai}'(at)/A, & u &= 2a \frac{\text{Bi}'(at) + i \text{Ai}'(at)}{B + iA}, & g_t(z) &= \frac{1}{2} - tz/2 - z^3/12. \end{aligned} \quad (182)$$

All complex logarithms in the following formulas are principal:

$$\begin{aligned} \Psi &= \frac{1}{2} g_t(v) p^2 - (t + v^2/2) p - v \log p + 2/(3p), \\ \Phi &= \frac{1}{2} g_t(u) p^2 + (t + u^2/2) p - u \log p - 2/(3p), \\ S(t) &= \frac{\Im \Psi}{\pi A^2}, \\ E(t) &= \frac{1}{\pi} \Im \left\{ \frac{\omega [4\bar{\Phi} - (4\pi i/3)\bar{u} - 2t^2]}{(A + iB)^2} \right\}, \quad \omega = e^{2\pi i/3}, \\ K(t) &= E(t) - S(t) - \frac{2t}{3a} + \frac{1}{a\sqrt{2t}}. \end{aligned} \quad (183)$$

Here $K(t)$ denotes a kernel and is unrelated to the entrance index K .

Theorem 23.1 (Real integral identity). *The integral in the following formula converges absolutely:*

$$c_1 = \alpha^{-1/6} \kappa \int_0^\infty K(t) dt. \quad (185)$$

The substitution $t = z^2$ gives a continuous integrand at zero,

$$2zK(z^2) = \frac{\sqrt{2}}{a} + 2z \left[E(z^2) - S(z^2) - \frac{2z^2}{3a} \right], \quad (2zK(z^2))_{z=0} = 2^{5/6}. \quad (186)$$

Proof. We include the branch and endpoint controls needed for the contour operations. Write $s = t - \lambda$, $A_s = \text{Ai}(as)$, $v_s = 2a \text{Ai}'(as)/A_s$, and

$$P(t, \lambda) = \frac{A_s w_\lambda(t)}{\text{Ai}(-a\lambda)} = \frac{\pi}{a} A_s [\text{Bi}(as) - r(\lambda) A_s], \quad H(t, \lambda) = P(t, \lambda) g_s(U_\lambda(t)).$$

The Wronskian gives

$$U = v_s + 2/P, \quad (\partial_t + \partial_\lambda)P = A_s^2 / \text{Ai}(-a\lambda)^2. \quad (187)$$

Polynomial division of the cubic gives

$$H = Pg_s(v_s) - s - v_s^2/2 - v_s/P - 2/(3P^2).$$

Therefore

$$\Psi_s(P) = \frac{1}{2}g_s(v_s)P^2 - (s + v_s^2/2)P - v_s \log P + 2/(3P) \quad (188)$$

is a primitive of H with respect to P , at fixed s .

The Green finite part. Let $Y = w^2 V_0$ and $b(\lambda) = a \text{Ai}'(-a\lambda) / \text{Ai}(-a\lambda)$. The Wronskian gives

$$L(t) = \frac{A_s}{\text{Ai}(-a\lambda)w(t)} = \int_t^\infty w(r)^{-2} dr, \quad L' = -w^{-2}.$$

Since $Y' = w^2 G$, integration by parts yields $I_0(T) = \text{FP} \int_0^T H dt - L(T)Y(T)$. At zero,

$$Y = -(2/3) \log t + \mathcal{O}(t^2), \quad L = t^{-1} + b + \mathcal{O}(t), \quad H = -2/(3t^2) - (2/3)b/t + 14\lambda/9 + \mathcal{O}(t).$$

These expansions define the finite part by adding $2/(3\delta) - (2/3)b \log \delta$ to the integral from δ . At infinity, $L(T)Y(T) = P(T)V_0(T) = -\sqrt{2s}/3 + o(1)$. Consequently

$$J_0(\lambda) = \lim_{T \rightarrow \infty} \left[\text{FP} \int_0^T H(t, \lambda) dt + (T - \lambda)^2/3 - \sqrt{2(T - \lambda)} \right]. \quad (189)$$

In particular, every finite constant is tied to (118); none is absorbed into an unspecified finite part.

Principal branches from resolvents. Put $C_+(s) = \text{Bi}(as) + i \text{Ai}(as) = 2e^{i\pi/6} \text{Ai}(aws)$ and $P_+(s) = (\pi/a)A_s C_+(s)$; define P_- by conjugation. For $\Im s < 0$, consider $\mathcal{L} = -d^2/dx^2 + x/2$ on $\mathbb{R}$ and spectral parameter $z = -s/2$. The right-decaying solution is $\text{Ai}(a(x+s))$, and the left-decaying one is $C_+(x+s)$. The connection formula and sector expansions show left decay as a power times $e^{-c_s \sqrt{|x|}}$, $c_s > 0$. Their Wronskian, in the order left then right, is $-a/\pi$. Thus the diagonal Green function is $P_+(s)$. Integrating its Green equation against its conjugate gives

$$\Im P_+(s) = \Im z \int_{\mathbb{R}} |G_s(x, 0)|^2 dx > 0. \quad (190)$$

The endpoint terms and weighted integrals converge by the stated decays. Hence P_+ stays in the upper half-plane throughout the lower s half-plane; its logarithm is principal, including both rotation sectors. Nonvanishing alone would not establish this branch assertion.

Similarly, $P(t, \lambda)$ is the diagonal Green function at t for $\mathcal{L}$ on $[0, \infty)$ with Dirichlet boundary at zero and spectral parameter $\lambda/2$: the Wronskian of w, A_s is $-\text{Ai}(-a\lambda)$. For $\lambda = i\tau$, $\tau > 0$, it has positive imaginary part. The left and right energy identities also give

$$\Re(1/P) \geq 1/t, \quad \Im(1/P) < 0, \quad |P| \leq t. \quad (191)$$

Indeed $1/P = (U - v_s)/2$, the left energy gives $\Re U \geq 2/t$, and the right decaying energy gives $\Re(-v_s) \geq 0$. Thus the actual P is in the first quadrant. All branches in the primitive are compatible; the lower half-axis is obtained by conjugation.

Normalized Fourier primitive. The raw primitive need not decay at both Fourier ends. Normalize each half:

$$L_{\pm}(t, \lambda) = \frac{\kappa}{A_s^2} [\Psi_s(P(t, \lambda)) - \Psi_s(P_{\pm}(s))], \quad \pm \Im \lambda > 0.$$

Because $(\partial_t + \partial_{\lambda})s = 0$, (187) gives $(\partial_t + \partial_{\lambda})L_{\pm} = \mathcal{F}H$. Let $\mathcal{I}$ denote upward imaginary-axis inversion at coordinate zero, $\mathcal{I}(R) = (2\pi i)^{-1} \int_{-i\infty}^{i\infty} R(\lambda) d\lambda$, and use L piecewise. The jump at zero is $L_{-}(t, 0) - L_{+}(t, 0) = 2i\kappa \Im \Psi_t(P_{+}(t))/A_t^2$. It follows that

$$\mathcal{I}(\mathcal{F}H) = \partial_t \mathcal{I}(L) + \kappa S(t). \quad (192)$$

We next justify its integrated endpoints rather than taking pointwise limits under the integral.

The origin endpoint uniformly in Fourier frequency. For $0 < t \leq 1$, Lemma 18.1 gives $|U(t, i\tau)| \leq B/t + B|\tau|t$, while $|v_{t-i\tau}| \leq B(1 + |\tau|)^{1/2}$. Together with (191) these give

$$|P|^{-1} \leq B(t^{-1} + 1 + |\tau|), \quad |\log P| \leq B + |\log t| + \log(1 + |\tau|).$$

Uniformly on $0 \leq t \leq 1$, outside a fixed compact frequency interval,

$$|A_{t-i\tau}|^{-2} \leq B(1 + |\tau|)^{1/2} e^{-c|\tau|^{3/2}}.$$

Substitution in (188) bounds the raw primitive divided by A_s^2 by

$$B(t^{-1} + 1 + |\log t|)(1 + |\tau|)^m e^{-c|\tau|^{3/2}} \quad (193)$$

for a fixed m . The reference primitive has the same kind of bound without the t^{-1} factor, since P_{+} and its inverse are polynomially bounded at infinity and nonzero on the remaining compact set. Split at $|\tau| = t^{-1/4}$. In the inner part, Taylor expansion gives

$$\frac{\Psi_{t-\lambda}(P(t, \lambda))}{A_{t-\lambda}^2} = \frac{1}{\text{Ai}(-a\lambda)^2} \left[\frac{2}{3t} - 2b(1 + \log t) \right] + \mathcal{O} \left(\frac{t(1 + |\tau|)^m(1 + |\log t|)}{|\text{Ai}(-a i\tau)|^2} \right). \quad (194)$$

For example, use $P = t + bt^2 + \mathcal{O}((1 + |\tau|)t^3)$, $v_s = 2b + \mathcal{O}((1 + |\tau|)t)$, and $A_s^{-2}/A_{-\lambda}^{-2} = 1 - 2bt + \mathcal{O}((1 + |\tau|)t^2)$. The small parameters $t\sqrt{1 + |\tau|}$ and $(1 + |\tau|)t^2$ are uniformly small on this interval. The integrated error is $\mathcal{O}(t(1 + |\log t|))$, using the Airy envelope. On the complementary interval, (193) is smaller than every fixed power of t after integration. The reference primitive converges by domination. Since $\mathcal{F}' = 2b\mathcal{F}$, $\mathcal{I}(\mathcal{F}b) = 0$. With $f_0 = f(0) = \kappa/a$ and

$$k_{\text{ref}} = \mathcal{I} \left(\frac{\kappa \Psi_{-\lambda}(P_{\pm}(-\lambda))}{\text{Ai}(-a\lambda)^2} \right),$$

we conclude

$$\mathcal{I}(L(t, \cdot)) = 2f_0/(3t) - k_{\text{ref}} + o(1). \quad (195)$$

The infinity endpoint including remote frequencies. Take $\tau \geq 0$, $s = T - i\tau$, $e(\tau) = r(i\tau) + i$, and $\zeta(s) = (\sqrt{2}/3)s^{3/2}$. Connection identities and the separate sector expansions give $|e(\tau)| \leq B e^{-c\tau^{3/2}}$ and, uniformly on $-\pi/2 \leq \arg s \leq 0$,

$$\begin{aligned} P_{+}(s) &= (2s)^{-1/2} [1 + \mathcal{O}(|s|^{-3})], \\ H_{+}(s) &= P_{+}(s) g_s(2C'_{+}(s)/C_{+}(s)) = -2s/3 + (2s)^{-1/2} + \mathcal{O}(|s|^{-2}). \end{aligned}$$

The exact relation $P = P_+(1 - d)$ has $d = e(\tau) \text{Ai}(as)/C_+(s)$. The global exponent difference satisfies

$$\begin{aligned} 2\Re[\zeta(T - i\tau) - \zeta(-i\tau)] &= \sqrt{2} \int_0^T \Re \sqrt{x - i\tau} dx \\ &\geq \int_0^T \sqrt{x + \tau} dx \geq \frac{T}{2\sqrt{2}} \sqrt{T + \tau}. \end{aligned}$$

Enlarging constants on bounded frequencies therefore gives $|d| \leq B e^{-cT\sqrt{T+\tau}} < 1/2$ for all sufficiently large T , uniformly in τ . The principal branches imply $\log P - \log P_+ = \log(1 - d)$ with no $2\pi i$ term. Taylor's theorem and $\Psi_s''(P) = g_s(v_s) + v_s/P^2 + 4/(3P^3) = \mathcal{O}(|s|^{3/2})$ give the uniform integrable expansion

$$L_+(T, i\tau) = -\frac{\kappa\pi}{a} e(\tau) H_+(T - i\tau) + \mathcal{O}((T + \tau)|e(\tau)|e^{-cT\sqrt{T+\tau}}). \quad (196)$$

The error integrates to $\mathcal{O}((T+1)e^{-cT^{3/2}})$. The remaining nonpolynomial part of $H_+ + 2s/3$ integrates to $\mathcal{O}(T^{-1/2})$ against $e(\tau)$. This covers $\tau \gg T$ as well as bounded τ without separating exponentially large factors. For $m_j = \mathcal{I}(\lambda^j \mathcal{F})$, integration by parts on each half-axis gives

$$\mathcal{I}(r - r_\pm) = \frac{a}{\pi\kappa} m_1, \quad \mathcal{I}(\lambda(r - r_\pm)) = \frac{a}{2\pi\kappa} m_2, \quad r_+ = -i, \quad r_- = i.$$

The origin boundary terms vanish because they are multiplied by λ or λ^2 . Hence

$$\mathcal{I}(L(T, \cdot)) = \frac{2}{3} T m_1 - \frac{1}{3} m_2 + \mathcal{O}(T^{-1/2}). \quad (197)$$

Fubini and the two finite parts. For each fixed $0 < \delta < T < \infty$, the energy bounds make H and the differentiated normalized primitive polynomial in $|\tau|$, while $\mathcal{F}$ supplies an exponential envelope. The integrations in (192) may therefore be exchanged before taking limits. For the upper endpoint in (189), define its truncated expression J_T . Splitting its t integral at $2(1 + |\tau|)$, the energy bounds on the initial part and the just-proved sector bounds on the remaining part give $|J_T(i\tau)| \leq B(1 + |\tau|)^m$ uniformly for $T \geq 1$. Indeed on the latter part $H + 2s/3 - (2s)^{-1/2} = \mathcal{O}(|s|^{-2})$ plus a damped connection term; if T is smaller than the splitting point, its endpoint terms are themselves polynomial in $|\tau|$. Thus dominated convergence against $\mathcal{F}$ is valid. Apply (192), (195), and (197) in (189). The elementary endpoint inversion is

$$f_0 T^2/3 - 2T m_1/3 + m_2/3 - f_0 \sqrt{2T} + o(1).$$

The $T m_1, m_2$ terms cancel exactly, and $f_0 = \kappa/a$ yields

$$\mathcal{I}(\mathcal{F} J_0) = k_{\text{ref}} + \kappa \int_0^\infty \left[S(t) + \frac{2t}{3a} - \frac{1}{a\sqrt{2t}} \right] dt. \quad (198)$$

This computation tracks both finite-part constants explicitly.

Rotation to positive real Airy arguments. In the lower s half-plane,

$$k_{\text{ref}}/\kappa = -\frac{1}{\pi} \Im \int_0^{-i\infty} \frac{\Psi_s(P_+(s))}{\text{Ai}(as)^2} ds.$$

Rotate from $\arg s = -\pi/2$ to $-2\pi/3$. The sector expansions give $P_+(s) \asymp s^{-1/2}$, $v_s = \mathcal{O}(|s|^{1/2})$, and $\text{Ai}(as)^{-2} = \mathcal{O}(|s|^{1/2} e^{-c|s|^{3/2}})$ throughout this closed wedge. The primitive quotient therefore

has an exponential envelope times a fixed polynomial and logarithm. There are no poles: Ai has no zeros there and the rotated Ai($a\omega s$) has argument in $[0, \pi/6]$. There is no singularity at zero. Both connecting arcs vanish. Equation (190) supplies the principal logarithm throughout, so no winding correction is omitted. At $s = \omega^2 t$, the connection identities give

$$\begin{aligned} \text{Ai}(a\omega^2 t) &= \frac{1}{2}e^{-i\pi/3}(A + iB), & C_+(\omega^2 t) &= 2e^{i\pi/6}A, \\ P_+(\omega^2 t) &= e^{i\pi/3}\bar{p}, & v_{\omega^2 t} &= \omega\bar{u}, \\ \log P_+(\omega^2 t) &= \overline{\log p} + i\pi/3, & \Psi_{\omega^2 t}(P_+) &= \omega[\bar{\Phi} - (i\pi/3)\bar{u}]. \end{aligned}$$

Here $0 < \arg p \leq \pi/6$, which also checks the displayed logarithm directly. Including $ds = \omega^2 dt$ gives

$$k_{\text{ref}}/\kappa = -\frac{1}{\pi}\Im \int_0^\infty \frac{4\omega[\bar{\Phi} - (i\pi/3)\bar{u}]}{(A + iB)^2} dt.$$

The same rotation for $m_2/2$ gives $m_2/(2\kappa) = -(1/\pi)\Im \int_0^\infty 2\omega t^2/(A + iB)^2 dt$. Their difference is precisely $\int E$. Since $q(0) = m_2/2 - \mathcal{I}(\mathcal{F}J_0)$, (198) proves (185). The following tail bounds prove its absolute convergence at infinity; at zero, E, S are finite because their Airy denominators and p are nonzero, proving (186). $\square$

24 An explicit tail bound and a certified sign

24.1 A positive real tail inequality

The following deliberately broad bound suffices to establish a strict sign:

$$\int_{10}^\infty |K(t)| dt < 0.141. \quad (199)$$

Here is a full derivation from bounded Airy remainders. Write $r = t^{-3/2}$, $\xi = (\sqrt{2}/3)t^{3/2}$ and define scaled functions by

$$\text{Ai}(at) = \frac{e^{-\xi}}{2\sqrt{\pi}(at)^{1/4}}x, \quad \text{Bi}(at) = \frac{e^\xi}{\sqrt{\pi}(at)^{1/4}}y, \quad \text{Bi}'(at) = \frac{(at)^{1/4}e^\xi}{\sqrt{\pi}}z.$$

The real remainder inequalities of [7] imply

$$\begin{aligned} x &= 1 - 5r/(24\sqrt{2}) + e_A r^2, & |e_A| &\leq 385/2304 < 0.1672, \\ y &= 1 + 5r/(24\sqrt{2}) + e_B r^2, & |e_B| &\leq 4(385/2304) < 0.6685, \\ z &= 1 - 7r/(24\sqrt{2}) + e_D r^2, & |e_D| &\leq 3(455/2304) < 0.5925. \end{aligned} \quad (200)$$

For Bi the relevant multiplier is $\chi(13/6) + 1 < \chi(3) + 1 = 3\pi/4 + 1 < 4$; for Bi' it is $\chi(2) + 1 = 3$, where $\chi(x) = \sqrt{\pi}\Gamma(x/2 + 1)/\Gamma(x/2 + 1/2)$ is increasing. The Ai and Ai' remainders use the first omitted term. For $t \geq 10$, $r < 0.032$, and these three scaled functions and the similarly scaled $-\text{Ai}'$ all belong to $[0.99, 1.01]$.

Set $p_0 = (\pi/a) \text{Ai}(at) \text{Bi}(at)$ and $U_B = 2a \text{Bi}'(at)/\text{Bi}(at)$. Then $p_0 = xy/\sqrt{2t}$, $U_B = \sqrt{2t}z/y$, and $H_\infty = p_0 g_t(U_B)$ satisfies the exact identity

$$\frac{H_\infty + 2t/3 - (2t)^{-1/2}}{t} = h(r, x, y, z) = \frac{2}{3} - \frac{xz}{2} - \frac{xz^3}{6y^2} + \frac{r(xy - 2)}{2\sqrt{2}}.$$

On the stated box direct differentiation gives $|h_x| < 0.70$, $|h_y| < 0.37$, $|h_z| < 1.04$. For the baselines x_0, y_0, z_0 obtained by deleting the remainders in (200), exact simplification gives

$$\frac{h(r, x_0, y_0, z_0)}{r^2} = -\frac{1875\sqrt{2}r^3 + 44680r^2 + 128256\sqrt{2}r + 1594368}{6912(5\sqrt{2}r + 48)^2}.$$

Its magnitude is below 0.101 on $[0, 0.032]$: bound the numerator at 0.032 and the denominator below by $6912 \cdot 48^2$. The mean-value bound is therefore

$$|h|/r^2 \leq 0.101 + 0.70(0.1672) + 0.37(0.6685) + 1.04(0.5925) = 1.081585 < 1.1.$$

Thus

$$|H_\infty + 2t/3 - (2t)^{-1/2}| \leq 1.1t^{-2} \quad (t \geq 10). \quad (201)$$

This is an explicit inequality, not a remainder guessed from a formal series.

To bound the difference from S , put $R = \text{Ai}(at)/\text{Bi}(at) > 0$. The real formula for S , obtained by taking the imaginary part in (183), is

$$S = \frac{1}{a} \left[g_t(v)p_0 - t - v^2/2 - \frac{v}{p_0} \frac{\arctan R}{R} - \frac{2}{3p_0^2(1+R^2)} \right]. \quad (202)$$

The Wronskian gives $U_B = v + 2/p_0$, and hence exactly

$$S - H_\infty/a = \frac{1}{a} \left[\frac{v}{p_0} \left(1 - \frac{\arctan R}{R} \right) + \frac{2}{3p_0^2} \frac{R^2}{1+R^2} \right].$$

Use $|1 - \arctan R/R| \leq R^2/3$ and the scaled bounds $p_0 \geq 0.69/\sqrt{t}$, $|v| \leq 1.5\sqrt{t}$, $R \leq 0.511e^{-2\xi}$. The resulting bracket is less than $3t$, and $3(0.511)^2 < 1$, so

$$|S - H_\infty/a| < te^{-(4\sqrt{2}/3)t^{3/2}}, \quad \int_{10}^{\infty} |S - H_\infty/a| dt < 10^{-20}. \quad (203)$$

For E , the same scaled bounds give

$$|p| \leq 0.74/\sqrt{t}, \quad |p|^{-1} \leq 1.45\sqrt{t}, \quad |u| \leq 1.5\sqrt{t}, \quad |\log p| \leq \log t.$$

For the last inequality, use $|\log p| \leq -\log |p| + |\arg p| \leq \frac{1}{2} \log t + \log 1.45 + 10^{-12} < \log t$. Therefore $|g_t(u)| \leq 1.05t^{3/2}$ and $|\Phi| \leq (2.83 + 1.5 \log t)\sqrt{t}$. In particular

$$4|\Phi - (i\pi/3)u| + 2t^2 \leq (17.64 + 6 \log t)\sqrt{t} + 2t^2 \leq 3t^2.$$

The final inequality follows at $t = 10$ from $17.64 + 6 \log 10 < 10^{3/2}$, and its gap is increasing. Also $1/(\pi|A + iB|^2) \leq 1.02\sqrt{t}e^{-2\xi}$. Consequently

$$|E(t)| \leq 3.1t^{5/2}e^{-(2\sqrt{2}/3)t^{3/2}}, \quad \int_{10}^{\infty} |E(t)| dt < 10^{-9}. \quad (204)$$

The logarithmic derivative of this majorant is at most -4 for $t \geq 10$, so its integral is at most one fourth of its value at 10. Combining (201)–(204) gives

$$\int_{10}^{\infty} |K(t)| dt \leq \frac{1.1}{10a} + 10^{-9} + 10^{-20} < 0.141,$$

proving (199).

24.2 Directed finite interval quadrature

The supplementary code evaluates (186) on $0 \leq z \leq 3.2$, so $0 \leq t \leq 10.24$. It uses only the Python standard library. Its arithmetic assumptions are stated explicitly: the documented correctly rounded Decimal operations [11] and the inspected program are trusted. This is not a proof-assistant formalization, an independent hardware verification, or a machine proof of the whole positive-real tail argument.

Intervals have directed lower and upper contexts at 70 decimal digits. Inputs are exact integers, rational numbers, or finite decimal strings; ordinary binary floats are rejected. Exp, logarithm, and square root are rounded to nearest and widened by adjacent representable numbers. General Decimal power is not used. Integer powers use directed multiplication. Reciprocals reject intervals containing zero; elementary functions reject inputs outside their proved domains. The positive arctangent helper rejects negative arguments, rather than silently invoking an unproved branch. Constructors and powers reject unsupported usages, including nonfinite values. Guards use explicit exceptions, so optimization with Python `-O` cannot remove them.

The Airy evaluator uses the convergent fundamental series for $f'' = xf$, with starting terms 1 and x . If the current term has degree n , its next term is multiplied by $x^3/((n+3)(n+2))$. Both value and derivative tails are bounded by geometric series once their explicit ratios are below one. It uses 64 steps on $0 \leq x \leq 9$, covering $az^2 \leq a(3.2)^2 < 9$. The initial values $\text{Ai}(0) = 1/(3^{2/3}\Gamma(2/3))$ and $\text{Ai}'(0) = -1/(3^{1/3}\Gamma(1/3))$ are enclosed using positive-real log-Gamma Stirling remainders [10]; the Bi values follow from the exact initial-value relations. The arithmetic starts with Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and exact rational alternating series remainders. For $x = 1/3, 2/3$, it evaluates $\log \Gamma(x + 100)$ with the eleven terms $B_{2k}/[2k(2k-1)(x+100)^{2k-1}]$, $1 \leq k \leq 11$, following the elementary Stirling main terms. The error lies between zero and the signed twelfth term. Subtracting $\sum_{j=0}^{99} \log(x+j)$ gives $\log \Gamma(x)$ before exponentiation. All Bernoulli numbers are generated as exact rationals. Endpoint monotonicity on $[0, \infty)$ encloses the Airy functions and first derivatives over each interval. Second derivatives and their chain rules follow directly from the Airy differential equation.

Second-order interval jets carry the value, first derivative, and second derivative of the transformed kernel through all operations. For an interval $[l, h]$ with exact midpoint m , its integral is enclosed by

$$(h-l)[2mK(m^2)] + \left[-\frac{(h-l)^3}{24}M_2, \frac{(h-l)^3}{24}M_2 \right], \quad M_2 \geq \sup_{l \leq z \leq h} |(2zK(z^2))''|. \quad (205)$$

This is Taylor's theorem integrated about the midpoint, not an estimate from successive floating quadratures. The real formula (202) avoids extracting an exponentially small imaginary part. Principal logarithms of first-quadrant numbers are evaluated by $\frac{1}{2} \log(x^2 + y^2) + i \arctan(y/x)$; arctangent is reduced by the half-angle identity before its alternating series remainder is enclosed.

The deterministic adaptive calculation starts with the 64 intervals of width $1/20$ covering $[0, 16/5]$ and bisects the interval of greatest enclosure width, with a fixed tie rule, until total width is at most 0.08. It produces 562 terminal panels. The mandatory replay regenerates them, checks their exact rational coverage and shared endpoints, recomputes every panel, and adds their directed enclosures independently of the adaptive accumulator. Thus the certificate is reproducible arithmetic, not a pasted numerical receipt. The resulting outward-rounded enclosures are

$$\int_0^{10.24} K(t)dt \in [-0.491128, -0.411312], \quad (206)$$

$$\alpha^{-1/6}\kappa \in [0.9540289883031181, 0.9540289883031182]. \quad (207)$$

The tail from 10.24 is bounded by the larger tail from 10 in (199). Using the full endpoints recorded in the replay gives

$$c_1 \in [-0.603067859400578, -0.257885944975658] \subset (-0.61, -0.25). \quad (208)$$

The enclosure in (208) is intentionally rounded outward. It completes the sign part of Theorem 15.1. It also proves $d = c_1/C < 0$ for the entire compact-positive-shift family. A sign certificate for this constant is not an effective error bound for approximating any particular finite coefficient.

25 Ancestor choices and local Gamma laws

Let $\mathcal{T}_n$ be the labeled trees on $\{0, 1, \dots, n\}$ rooted at 0, and let $d_T(v)$ be the distance from the root. The exact identity is

$$P_n(y) = \sum_{T \in \mathcal{T}_n} \prod_{v \neq 0} (y + d_T(v) - 1). \quad (209)$$

Indeed the root has an unordered labeled set of branches. A branch root contributes xy , and its descendants have generating function F_{y+1} ; the labeled exponential formula is precisely (111). The finite tower restricts maximum depth, explaining the exact-height coefficients in (123). Formula (209) also proves that each P_n has nonnegative integer coefficients.

Decorate each nonroot vertex v by choosing one of its $d_T(v)$ strict ancestors. Give its choice of root 0 weight y , and each other ancestor choice weight 1. Summing decorations gives exactly the product in (209). Choose uniformly from all such decorated trees at $y = 1$, and let K_n count vertices that chose root 0. Its probability generating function is $P_n(y)/P_n(1)$.

The universal relative correction cancels when the two coefficient expansions are divided. Thus

$$\mathbb{E}y^{K_n} = \frac{(n/\rho)^{y-1}}{\Gamma(y)} [1 + o_Y(n^{-1/3})] \quad (210)$$

uniformly for compact positive real y sets. For real fixed t take $y = e^{t/\sqrt{L_n}}$, $L_n = \log(n/\rho)$. Then

$$\log \mathbb{E}e^{t(K_n - L_n)/\sqrt{L_n}} = -t\sqrt{L_n} + L_n(e^{t/\sqrt{L_n}} - 1) - \log \Gamma(e^{t/\sqrt{L_n}}) + o(1) \rightarrow t^2/2.$$

The real MGF continuity theorem proves the normal limit in (114). No differentiation of an error term is needed for it.

25.1 Local complex control and fixed cumulants

A rate on a real interval plus positivity gives a further local result. Let γ be Euler's constant and let ζ denote the Riemann zeta function.

Theorem 25.1 (Local Gamma correction to the Poisson transform). *Uniformly for complex $|z - 1| \leq 1/16$,*

$$\mathbb{E}z^{K_n} = \frac{e^{L_n(z-1)}}{\Gamma(z)} [1 + \mathcal{O}(n^{-5/24})]. \quad (211)$$

For every fixed positive integer r , the ordinary cumulant satisfies

$$\text{cum}_r(K_n) = L_n + \gamma - \sum_{k=2}^r \left\{ \begin{matrix} r \\ k \end{matrix} \right\} (-1)^k (k-1)! \zeta(k) + \mathcal{O}_r(n^{-5/24}), \quad (212)$$

where the braces are Stirling numbers of the second kind. In particular,

$$\mathbb{E}K_n = L_n + \gamma + \mathcal{O}(n^{-5/24}), \quad \text{Var } K_n = L_n + \gamma - \zeta(2) + \mathcal{O}(n^{-5/24}). \quad (213)$$

Proof. On $|z - 1| \leq 1/4$ define the analytic residual

$$R_n(z) = \Gamma(z)e^{L_n(1-z)}P_n(z)/P_n(1) - 1.$$

There are no Gamma poles on this disk. On its real diameter, (210) gives $|R_n(y)| \leq C_0 n^{-1/3}$ for sufficiently large n , uniformly in $y \in [3/4, 5/4]$. For the outer complex bound, positivity gives $|P_n(z)| \leq P_n(|z|)$. The leading real family estimate and bounded compact Gamma factors therefore give

$$|R_n(z)| \leq C_1 n^{|z| - \Re z} + 1 \leq C_2 n^{1/24}, \quad |z - 1| \leq 1/4, \quad (214)$$

since $\Re z \geq 3/4$, $|\Im z| \leq 1/4$, and $|z| - \Re z = (\Im z)^2/(|z| + \Re z) \leq 1/24$. In particular a bounded outer normal family is not assumed.

Set $w = 4(z - 1)$ and work in the upper half unit disk. The harmonic measure of its semicircular arc is

$$v(w) = \frac{2}{\pi} \Im \log \frac{1+w}{1-w}.$$

The quotient has positive real and imaginary parts in the half disk, so the logarithm is unambiguous. Its boundary values are 0 on the open diameter and 1 on the open arc. At fixed $|w|$, the argument is maximal at $w = i|w|$, because its tangent is $2\Im w/(1 - |w|^2)$. Hence for $|w| \leq 1/4$,

$$0 \leq v(w) \leq (4/\pi) \arctan(1/4) < 1/3.$$

Apply the subharmonic maximum principle to $\log |R_n|$, using the real-diameter bound and (214). With $C = \max(C_0, C_2)$ it gives

$$\log |R_n(z)| \leq \log C + \left[-\frac{1}{3}(1 - v(w)) + \frac{1}{24}v(w) \right] \log n \leq \log C - \frac{5}{24} \log n.$$

Zeros of the residual are allowed for this subharmonic function. At the two corners, continuity of R_n and the stronger endpoint real bound give nonpositive upper limits for its difference from the harmonic majorant; alternatively their harmonic measure is zero. Reflection gives the lower half disk, and the diameter already has the stronger bound. This proves (211).

For large n , $|R_n| < 1/2$ on the smaller disk, so $\log(1 + R_n)$ is analytic and $\mathcal{O}(n^{-5/24})$ there, normalized to zero at $z = 1$. Gamma also has a normalized analytic logarithm there. For $|t| \leq 1/32$, $|e^t - 1| \leq e^{1/32} - 1 \leq 1/31 < 1/16$. Therefore

$$\log \mathbb{E} e^{tK_n} = L_n(e^t - 1) - \log \Gamma(e^t) + \log(1 + R_n(e^t))$$

is the analytic cumulant-generating function, with an analytic uniformly $\mathcal{O}(n^{-5/24})$ error. Cauchy's formula bounds its r th error derivative by $r!32^r$ times this supremum for each fixed r . Finally, the Euler-operator identity gives

$$\left. \frac{d^r}{dt^r} \log \Gamma(e^t) \right|_{t=0} = \sum_{k=1}^r \left\{ \begin{matrix} r \\ k \end{matrix} \right\} (\log \Gamma)^{(k)}(1).$$

The convergent Gamma derivative identities [10] are $(\log \Gamma)'(1) = -\gamma$ and $(\log \Gamma)^{(k)}(1) = (-1)^k(k - 1)!\zeta(k)$ for $k \geq 2$. Subtracting this contribution proves (212) and (213). $\square$

For example the third constant is $\gamma - 3\zeta(2) + 2\zeta(3)$ and the fourth is $\gamma - 7\zeta(2) + 12\zeta(3) - 6\zeta(4)$. The exponent $5/24$ is deliberately nonoptimal. This theorem is local near $z = 1$; it is not a global complex- y estimate, a lattice local limit for K_n , control of cumulants with growing order, or an explicit finite- n error certificate. The finite-height local limit in (173) concerns a different random variable.

26 The positive real Borel boundary

Let $B(z) = \sum_{n \geq 0} b_n z^n$. The coefficient theorem implies that its radius is ρ and, as real $z \uparrow \rho$,

$$B(z) = \sqrt{\frac{2}{\rho}}(1 - z/\rho)^{-1/2} + c_1 \Gamma(1/6)(1 - z/\rho)^{-1/6} + o((1 - z/\rho)^{-1/6}). \quad (215)$$

To prove this, put $x = z/\rho$ and $s = -\log x$. For $0 < \beta < 1$, integral comparison for the positive decreasing function $t^{-\beta} e^{-st}$ gives

$$\sum_{n \geq 1} n^{-\beta} e^{-sn} = \Gamma(1 - \beta) s^{\beta-1} + \mathcal{O}(1).$$

Apply this at $\beta = 1/2$ and $5/6$. Replacing $s = (1 - x) + \mathcal{O}((1 - x)^2)$ by $1 - x$ changes neither displayed term at the remainder scale. For a sequence $r_n = o(n^{-5/6})$, split the generating sum at a fixed N ; its finite prefix is $\mathcal{O}(1)$ and its tail is bounded by $\eta \sum n^{-5/6} x^n$ for all sufficiently large N . Letting $\eta \downarrow 0$ gives the asserted little- o . Finally $C\Gamma(1/2) = \sqrt{2/\rho}$. This is a one-sided real Abelian conclusion. No complex sector or Δ -domain Borel singular expansion is asserted.

27 Threshold inversion at the fractional scale

Write

$$\nu_y = y - \frac{1}{2}, \quad C_y = C\rho^{1-y}/\Gamma(y), \quad d = c_1/C, \quad M_y(t) = 2t \log \frac{t}{e\sqrt{\rho}} + \nu_y \log t + \log(2\pi C_y).$$

The squared Stirling formula and Theorem 15.2 give

$$\log P_n(y) = M_y(n) + dn^{-1/3} + o_Y(n^{-1/3}) \quad (216)$$

uniformly on compact positive Y . There is an exact useful lower bound: restrict the root to a single child in (209), then use coefficient positivity in the shift to obtain

$$P_n(y) \geq nyP_{n-1}(y+1) \geq nyP_{n-1}(y). \quad (217)$$

Thus $P_n(y)$ is strictly increasing once $ny > 1$, with a common eventual index for $y \in Y$. Let g_y be the piecewise-linear interpolation of $\log P_n(y)$ on that eventually increasing part. For sufficiently large L define $z_y(L) = g_y^{-1}(L)$. The finitely many earlier values are bounded uniformly on Y , so they cannot create an earlier crossing of a large threshold. Hence exactly

$$N_y(Y_0) := \min\{n \geq 0 : P_n(y) \geq Y_0\} = \lceil z_y(\log Y_0) \rceil \quad (218)$$

for sufficiently large Y_0 .

Let $x_y = x_y(L)$ be the unique large positive solution $M_y(x_y) = L$. Its derivative is $M'_y(t) = 2 \log(t/\sqrt{\rho}) + \nu_y/t$, positive uniformly on Y for large t . Interpolation of M_y contributes $\mathcal{O}_Y(1/t)$ because $M''_y(t) = 2/t - \nu_y/t^2$; interpolation of $dt^{-1/3}$ and the uniform little- o sequence preserves the error in (216). First bounding $z_y - x_y$ by $\mathcal{O}_Y(x_y^{-1/3}/\log x_y)$ and then Taylor expanding therefore proves

$$z_y(L) = x_y - \frac{dx_y^{-1/3}}{2 \log(x_y/\sqrt{\rho})} + o_Y(x_y^{-1/3}/\log x_y). \quad (219)$$

Replacing M'_y by the displayed denominator changes only a lower-order term. Since $d < 0$, the fractional term increases the inverse relative to the smooth leading-log inverse.

27.1 An explicit common Lambert base

For the positive real Lambert function W_0 , put

$$X = \frac{L}{2W_0(L/(2e\sqrt{\rho}))}, \quad D = 2\log(X/\sqrt{\rho}), \quad d_y = \frac{\nu_y \log X + \log(2\pi C_y)}{D}, \quad r_y = X - d_y. \quad (220)$$

Here d_y is a Newton displacement, distinct from the universal d . The number X solves $f(X) = L$ for $f(t) = 2t \log(t/(e\sqrt{\rho}))$; $f'(X) = D$ and $f''(t) = 2/t$. Uniformly on Y , $d_y = \mathcal{O}_Y(1)$, and Taylor's theorem gives explicitly

$$M_y(r_y) - L = (d_y^2 - \nu_y d_y)/X + \mathcal{O}_Y(X^{-2}).$$

The linear cancellation is exact because $Dd_y = \nu_y \log X + \log(2\pi C_y)$. It follows that $x_y - r_y = \mathcal{O}_Y(1/(X \log X))$, which is smaller than $X^{-1/3}/\log X$. Replacing x_y by X in the fractional correction also changes only a smaller term. Consequently

$$z_y(L) = r_y - \frac{dX^{-1/3}}{2\log(X/\sqrt{\rho})} + o_Y(X^{-1/3}/\log X). \quad (221)$$

This accuracy is established by the controlled Newton residual, not inferred from a coarser leading inverse. At $y = 1$ the expression simplifies to

$$z_1(L) = X - \frac{1}{4} - \frac{\log(8\pi/\sqrt{\rho})}{4\log(X/\sqrt{\rho})} - \frac{(c_1/C)X^{-1/3}}{2\log(X/\sqrt{\rho})} + o(X^{-1/3}/\log X). \quad (222)$$

Let $R_y(L)$ denote the displayed approximation in (221). For each compact Y there is a positive $\eta(L) \rightarrow 0$ such that, uniformly for $y \in Y$,

$$\left| R_y(L) - \eta(L)X^{-1/3}/\log X \right| \leq N_y(e^L) \leq \left| R_y(L) + \eta(L)X^{-1/3}/\log X \right|. \quad (223)$$

It follows from (218) and the uniform real inverse error. This is a shrinking-width asymptotic sandwich, not a computable error certificate. A universal single ceiling of R_y is not justified: thresholds can lie arbitrarily close to integer crossings.

[write] 7 October 2026. (218) is an instance of item (1) of the transseries volume's `p0:thm:staircase`, and (223) follows from it as item (2) does; X is an instance of `p0:prop:factorial-core`; the Newton and fractional corrections are this Part's own (front matter, "The inverses and the transseries volume"). At $y = 1$, (222) is Part I's $r(Y)$ of (14) with the fractional term added.

28 Reproducibility and source scope

The supplementary archive contains the editable TeX, the PDF, exact finite checks, the directed interval certificate, regression fixtures, deterministic receipts, and an offline builder. The mathematical proofs are in this article. Finite checks support identities and normalizations; they do not prove an infinite asymptotic statement by testing examples.

The mandatory replay checks polynomial rows through degree 30; independently enumerates 18,248 Prüfer trees through six nonroot vertices; checks six positive rational shifts against independent finite-tower expansions; and checks 17 rational inequalities used after the analytic tail estimates. A separate arithmetic test suite performs 20,008 deterministic inclusion and normalization checks. Domain and file-integrity guards have 267 tests. Every test uses explicit failure conditions that

remain active with Python -0. The certificate regenerates and independently reaggregates all 562 terminal panels in both modes. The panel verifier is independent of adaptive scheduling and running-total bookkeeping, but reuses the interval kernel and panel implementation; it is not a second implementation of the entire numerical method.

The archive documents exact commands. A code-only run needs Python 3.12 or later and its standard library; it requires no network, special-function package, or TeX. The full builder additionally needs an installed pdfLaTeX stack. It checks the source inventory, runs the mandatory replay, compiles with shell escape disabled and fixed metadata, and publishes only into a fresh output directory. The ZIP has a sorted explicit inventory and fixed entry metadata. Byte identity is intended for the same source and installed Python/TeX stack; cross-version PDF identity is not promised. Existing outputs, symlinked paths, unexpected files, and altered fixtures are rejected. No private research paths, timing logs, caches, or downloaded paper bodies are included.

Hashes establish integrity relative to the received bundle; they do not authenticate an adversary's replacement of both code and hashes. The Python implementation, operating system, and source directory are trusted. The mathematical and arithmetic trust boundaries are distinct: the sign certificate depends on the proved real integral and tail inequality, and the interval code does not machine-formalize those proofs. Optional ordinary floating diagnostics are not part of the mandatory certificate or its inputs.

[write] 7 October 2026. In this report the package's programs are `code/193-fraction-*.py`, its fixtures and provenance data `data/193-fraction-code-*`, its generated outputs `data/193-fraction-generated-*`, and its `README_REPRODUCIBILITY.md`, `DATA_SOURCES.md`, `code/README.md` and `code/fixtures/README.md` are `193-fraction-*.md`. `Report193.tex` (its text is this Part), the delivered PDF, the package `README.md`, `SHA256SUMS.json` and the three files `generated/certificate/coefficient.json`, `core.json` and `panels.json`, byte copies of the fixtures `code/fixtures/*_reference.json`, are not shipped. The programs find their fixtures and `PROVENANCE.json` by their delivered paths; the report's `README` rebuilds that layout on a copy.

28.1 Attribution and the limit of the literature comparison

Hanna's OEIS entries explicitly record both the unshifted sequence and the entire shifted equation (111). The broad continued-exponential and weighted-tree framework is prior work of Bender and Vinson. Their 1996 paper and Poland's 1998 continuation remain relevant; full-text theorem overlap has not been resolved, so no search failure is used as evidence of novelty. The present claims concern the displayed theorems and proofs, with those attributions retained.

The retrieved A096542 entry has an indexing inconsistency in one formula line: it states $[y]P_n(y) = na_n$, whereas its rows and the formal equation give $[y]P_n(y) = na_{n-1}$. Indeed taking the coefficient of y at $y = 0$ in $F_y = e^{xyF_{y+1}}$ gives $[y]F_y = xF_1$. For example $[y]P_3(y) = 15 = 3a_2$. The reproducible checks use the defining formal equation and this independently derived indexing, not the inconsistent line.

[write] 7 October 2026. Part I records the same mismatch (Section 13); Remark 13.1 states the corrected formula with its proof and checks it against the entry's rows.

The primary analytic references are the standard Airy Wronskians, connection formulas, sector expansions, real remainder inequalities, and Gamma identities in the NIST DLMF. The discrete entrance, outer error, Fourier transfer, Green finite-part reduction, explicit tail estimates, and local complex continuation argument have been supplied here. There is no all-orders coefficient theorem, global complex row-PGF theorem, row-statistic local limit, effective asymptotic onset, or complex Borel-sector expansion in this report. Such statements require additional analysis.

28.2 Directions for further analysis

The estimates suggest several concrete next questions, which remain open here.

1. **The next coefficient scale.** Derive and justify the next modified Riccati equation and outer corrector, retaining any logarithmic entrance terms until after height averaging. This requires a new overlap budget and a second-order growing-window Fourier remainder. The present $\epsilon^{17/16}$ error does not justify a further coefficient.
2. **Sharper certified constants and finite-index control.** A finer finite quadrature and additional rigorously bounded Airy tail terms could narrow the interval for c_1 . An effective onset for approach from below would separately require explicit constants throughout the discrete remainder proof; improving the constant's decimal enclosure alone is insufficient.
3. **Varying shifts and broader probability laws.** The regimes $y \downarrow 0$ and $y \rightarrow \infty$ require uniform treatment of the lower endpoint and Gamma factor. A local limit for K_n would require control of its full Fourier circle, beyond the local complex disk proved here. The existing finite-height minor arcs concern $N_{h,y}$ and cannot be transferred to K_n without an additional argument.
4. **Complex Borel continuation and effective inversion.** A sectorial Borel expansion needs analytic continuation and complex bounds, not just the positive-real Abelian estimate. A certified integer-threshold algorithm would need an effective coefficient remainder and explicit crossing tests before replacing a shrinking ceiling sandwich by a definite integer output.
5. **The full prior-work comparison.** A page-level comparison with authorized full texts of Bender–Vinson and Poland would clarify the overlap in unbounded linear weights, critical finite depth and coefficient asymptotics. Until that comparison is complete, priority remains open.

[write] **7 October 2026.** These directions are gathered with Part I's questions in Section 29.

29 Further questions and research

[Added 7 October 2026, batch 110.] This section gathers, under Vladimir's standing rule of 4 October 2026, the open questions of both manuscripts: Part I's five "Questions not settled by the present estimates" (Section 14) and Part II's five "Directions for further analysis" (Section 28), with what Part II settles of Part I's list. The delivered lists stay where they are. No claim of either manuscript was found to be false; the two refuted statements are OEIS formula lines (Remarks 1.4 and 13.1).

1. *Rates and further coefficients* (Part I, item 1; Part II, item 1). Part II proves the first correction, $b_n \rho^n \sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3})$ (Theorem 15.1), which settles the first-coefficient part of Part I's item. Open: an effective rate, and the next coefficient scale, which needs a further modified Riccati equation and outer corrector, a new overlap budget and a second-order growing-window Fourier remainder; Part II says its $\epsilon^{17/16}$ error "does not justify a further coefficient".
2. *Fixed positive shifts* (Part I, item 2; Part II, item 3). Part II proves the Gamma prefactor and the same relative correction uniformly on compact sets of shifts $y > 0$ (Theorem 15.2), which answers Part I's item for every fixed positive shift. Open: uniformity as $y \downarrow 0$ or $y \rightarrow \infty$, and a local limit for the root-ancestor statistic K_n , which would need control of its full Fourier circle.

3. *Larger tilt domains and rare heights* (Part I, item 3). The real Airy window stops at the convenient bound $\lambda \leq 1$; extending it toward the Airy zeros, and large-deviation equivalents for the height away from the central range, need further analysis. Open.
4. *Effective inversion, sharper certified constants and onsets* (Part I, item 4; Part II, items 2 and 4). Both inverse sandwiches are asymptotic existence statements. A certified threshold algorithm needs explicit constants and onsets for the coefficient error and a crossing-aware integer comparison; a narrower enclosure of c_1 needs finer quadrature and more rigorously bounded Airy tail terms, and an effective onset of the approach from below needs explicit constants throughout the discrete remainder proof. Open.
5. *Complex Borel continuation* (Part II, item 4). Part II's Borel boundary (215) is a one-sided real Abelian statement; a sectorial expansion needs analytic continuation and complex bounds. Open.
6. *The full prior-work comparison* (Part I, item 5; Part II, item 5). A page-level comparison with the full texts of Bender–Vinson and Poland, on unbounded linear weights, critical finite depth and coefficient asymptotics. Until then neither Part claims priority; the write did not read those papers either. Open.

References

- [1] P. D. Hanna, sequence A096537, *The On-Line Encyclopedia of Integer Sequences*, created 24 June 2004; includes the amplitude conjecture attributed to V. Kotešovec, 16 December 2019. <https://oeis.org/A096537>. Retrieved 4 October 2026.
- [2] P. D. Hanna, triangle A096542, *The On-Line Encyclopedia of Integer Sequences*, created 25 June 2004. <https://oeis.org/A096542>. Retrieved 4 October 2026.
- [3] C. M. Bender and J. P. Vinson, Summation of power series by continued exponentials, *Journal of Mathematical Physics* **37**(8) (1996), 4103–4119. <https://doi.org/10.1063/1.531619>. Primary metadata and abstract inspected; full-text theorem scope unresolved.
- [4] D. Poland, Summation of series in statistical mechanics by continued exponentials, *Physica A* **250** (1998), 394–422. [https://doi.org/10.1016/S0378-4371\(97\)00533-5](https://doi.org/10.1016/S0378-4371(97)00533-5). Primary metadata and abstract inspected; full-text theorem scope unresolved.
- [5] F. Chapoton, Free hyperplane arrangements associated to labeled rooted trees (2003). Preprint arXiv math/0301372, version 2; see Propositions 6 and 7. <https://arxiv.org/abs/math/0301372>.
- [6] NIST Digital Library of Mathematical Functions, Section 9.2, Airy differential equation, initial values, Wronskian and connection formulas; especially equations 9.2.3, 9.2.7, 9.2.10 and 9.2.12. <https://dlmf.nist.gov/9.2>.
- [7] NIST Digital Library of Mathematical Functions, Section 9.7, Airy asymptotic expansions; especially equations 9.7.5–9.7.8, including the separate function and derivative expansions. <https://dlmf.nist.gov/9.7>. Report 193 cites the same section for the real and complex error bounds.
- [8] NIST Digital Library of Mathematical Functions, Section 9.9(i), location of the zeros of the Airy function. <https://dlmf.nist.gov/9.9>.
- [9] *Report192: Airy Limits and the Corrected Asymptotic of A096537*, 4 October 2026. Companion leading-result report. The present report states its leading facts explicitly and supplies the quantitative estimates required for the sequel. [Printed as Part I of this report.]
- [10] NIST Digital Library of Mathematical Functions, Sections 5.2, 5.7 and 5.11, Gamma function, convergent psi series, and real Stirling remainders. <https://dlmf.nist.gov/5.2>; <https://dlmf.nist.gov/5.7.E4>; <https://dlmf.nist.gov/5.11.ii>.
- [11] Python Software Foundation, *Decimal fixed-point and floating-point arithmetic*, Python standard-library documentation. <https://docs.python.org/3/library/decimal.html>.
- [12] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, the transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; Part 0, `p0:thm:staircase`, `p0:prop:factorial-core` and `p0:prop:operator-form`. Cited by the merge (batch 110).